When Design Exploration Gets Cheap

Upstream Search, Downstream Truth, and the Migration of Scarcity in Chip Design

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Brad Pierce developed the central formulation and retains final editorial responsibility. AI assisted in preparing the paper. This collaboration does not imply endorsement by any organization.

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Abstract

When AI lowers the cost of design exploration faster than it lowers the cost of authoritative evaluation, chip design reorganizes around what becomes relatively scarce: trusted evidence, accumulated context, engineering memory, and ultimately the authority to commit. The allocation problem is evaluator economics: deciding what evidence to obtain next, which evaluator has enough authority for the question at hand, when stronger evidence is worth its cost, and when to stop or escalate. EDA makes this evaluator-allocation problem unusually visible because its evaluators differ sharply in cost, latency, coverage, fidelity, authority, and consequence. The same cost asymmetry reorganizes control and value. Who controls the workflow, whose capability is difficult to substitute, and where economic value accrues are distinct questions and can have different answers. State, methodology, failure history, routing knowledge, and qualified integration can accumulate into technical capital even when they never become standalone software products. Separately owned technical capital can also operate as one engineering system when governed interfaces preserve authority, provenance, qualification, and ownership while allowing external evaluators to contribute authoritative evidence. Public evidence already shows the layered architecture in real deployment. That leaves four linked questions: where scarcity moves as exploration becomes cheap, what capital forms around the new scarcities, which ownership structures those conditions favor, and which of those assets can support durable economic value.

Keywords: electronic design automation, EDA, artificial intelligence, agentic AI, evaluator economics, routing capital, technical capital, technical capital intelligence, techcap, semiconductor design, design-space exploration

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Executive Summary

Chip design is entering an asymmetric regime: AI is reducing the cost of proposing, exploring, and revising alternatives faster than it's reducing the cost of authoritative engineering evidence. Moving intelligence above mature EDA engines doesn't make those engines dispensable. A machine actor can generate candidates, invoke tools, interpret results, remember outcomes, and try again while trusted engines continue to settle different engineering questions. Throughout this report, "truth" is shorthand for engineering evidence with enough domain-specific authority to settle a particular question under the relevant assumptions. [3, 7, 13, 15] As a deliberately informal shorthand, I call the cost of obtaining that question-settling evidence "dollars-per-truth." As adjacent work gets cheap, scarcity can move toward whatever AI hasn't made easier.

As alternatives get cheaper, deciding what evidence to buy next matters more. Evaluator economics asks which evaluator to run, when stronger evidence is worth its cost, when the evidence is sufficient, and when to stop or escalate. EDA makes the problem unusually visible because evaluators differ sharply in cost, latency, coverage, fidelity, authority, and consequence; authority isn't the same as fidelity. Cheap proxies can widen search while stronger checks are invoked selectively as escalation points. [19, 185, 186, 187] The evaluator-economics framing draws on value-of-information analysis, value-of-computation analysis, engineering design decision analysis, multi-fidelity optimization, and risk-informed verification and validation. [22, 23, 24, 25, 51, 67, 188, 189] If evaluation also gets cheap, scarcity can move again toward commitment.

AI can take on more of the work above and between mature engines: proposing work, routing it, interpreting evidence, retaining state, and choosing the next experiment. It can also make automation approaches that used to be limited by human authoring, coordination, interpretation, or search effort more practical. AI can relax those constraints. It does much less for QoR limits, solver complexity, fidelity limits, process dependence, or physics. [9, 13, 14] Control, dependence, and economic capture are distinct questions: resource dependence shapes bargaining power, while complementary assets shape who captures value. [65, 77] Public evidence already shows incumbent-provider, design-organization, independent-provider, and hybrid arrangements. [3, 5, 6, 15, 18]

What underlies the agent can matter more than the visible interface: current design state, accepted constraints, provenance, evaluator results, methodology, experiment history, failure history, and authority boundaries. Engineering memory and routing policy become technical capital only when lessons remain tied to the conditions that made them valid. That capital can accumulate inside a design organization, an incumbent platform, a governed multi-party system, open infrastructure, or national capability; as workflows and constraints change, its strategic importance can shift even when it never becomes standalone software revenue. Who ultimately captures the value remains open.

The clearest publicly quantified commercial evidence already establishes real deployment: Synopsys reported its first 100 commercial DSO.ai tape-outs by February 2023; Cadence reports more than 1,000 Cerebrus tapeouts and expanding ChipStack deployments; and ChipAgents reports more than 200 semiconductor-enterprise deployments, with named workflow results from Andes, eMemory, Whalechip, and Ambiq. [1, 2, 3, 6, 145, 146, 147, 148, 149]

Those facts don't establish long-run switching cost, retention, deployment burden, cross-organization ownership of engineering memory, or category economics. The technical architecture is more visible than the eventual ownership and market structure. The business questions are where scarcity moves, what accumulates around it, which actors can capture or retain those assets under current conditions, how transferable those assets are, and which assets can support durable economic value.

Evidence state	Meaning	Where the report stands
SUPPORTED	Public evidence is sufficient within the stated scope. Company-reported metrics count for what they show, not more: deployment doesn't prove retention, switching cost, or durable economics.	Agentic systems can operate above trusted EDA engines. AI-driven optimization and agentic verification have entered real enterprise deployment. Differences in evaluator cost and authority strongly shape how much autonomy is practical. Different parts of the loop can be controlled by incumbent providers, design organizations, independent providers, or hybrid arrangements. Cross-design reusable engineering skill is technically demonstrated. Different ecosystems converge on similar architectures linking search, trusted evaluation, evidence and state, and the next experiment.

Evidence state	Meaning	Where the report stands
CONDITIONAL / SCENARIO	The mechanism or direction is credible, but public evidence doesn't justify a categorical conclusion.	Engineering memory creates a durable vendor advantage. A vendor-neutral orchestration layer becomes a major control point. Higher-level representation vendors become large standalone markets. Agentic AI raises total EDA industry spending. Usage-based pricing broadly displaces seat licensing. The right to commit becomes a major organizational constraint as search scales.
OPEN — PUBLIC EVIDENCE INSUFFICIENT	Public evidence is insufficient to conclude. OPEN means unresolved, not false; decisive facts may be private or unobservable.	Private procurement and build-versus-buy logic. Renewal, churn, and net retention for new agentic vendors. Deployment-services burden. Cross-organization ownership of persistent engineering memory. Industry-wide total-cost elasticity. Detailed commercial dollars-per-truth curves across evaluator classes.

1. The New Economics of Search and Evidence

1.1 The Stack Is Asymmetric

Electronic design automation is often discussed as if it were a single stack that AI either replaces or fails to replace. That framing hides the most important asymmetry.

At the top of the design process, the problem is still plastic. Architecture can change. Microarchitecture can change. Numerical representation can change. Work can move between hardware and software. Memory organization, pipelining, parallelism, interfaces, and control structure can still be reconsidered. Higher-level representations can generate families of implementations instead of one fixed RTL description.

Farther downstream, the design becomes progressively entangled with physical reality. Timing closure, congestion, parasitics, power integrity, clock distribution, process corners, analog behavior, manufacturing rules, design-rule checking and layout-versus-

schematic verification (DRC/LVS), electromigration and IR-drop analysis (EM/IR), packaging, libraries, and foundry qualification become increasingly expensive to approximate and costly to get wrong.

The asymmetric design flow produces two opposing gradients. Upstream, optionality and reversibility are high; downstream, the cost and authority of truth rise.

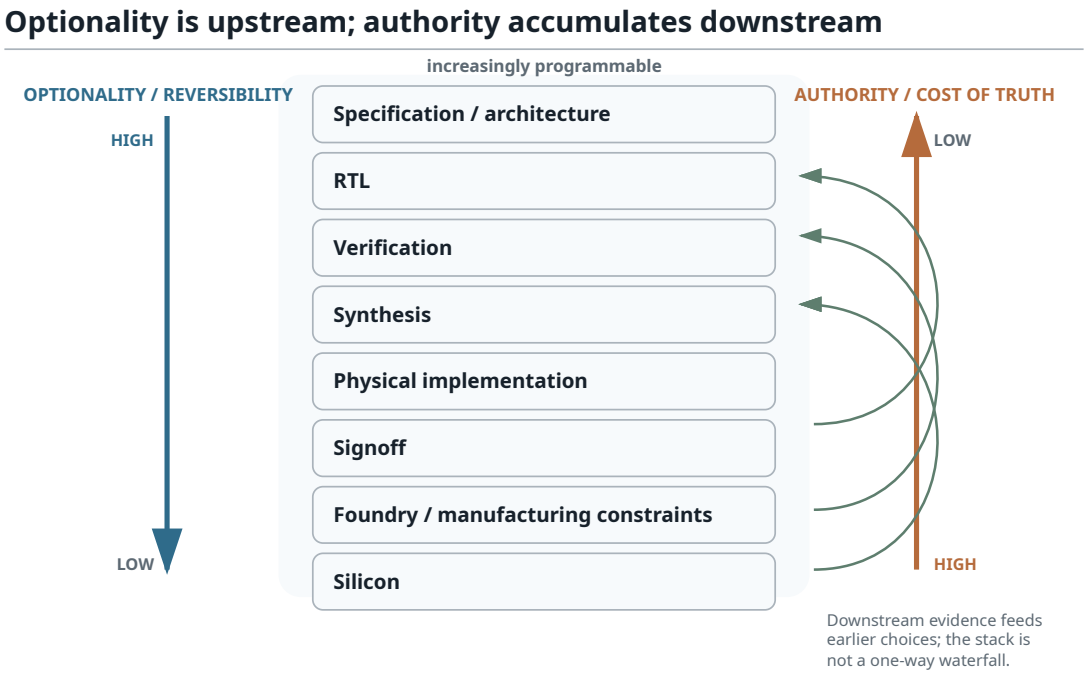


Figure 1. AI can make proposal and exploration dramatically cheaper while reducing the cost of authoritative physical and manufacturing evidence much less. The resulting opportunity is asymmetric: search can move rapidly without similarly dispersing qualification, process integration, and trusted downstream evaluation.

AI changes upstream and downstream search economics differently; calling one “easy” and the other “hard” hides the mechanism.

A synthesis engine can optimize a given RTL description extremely well. A place-and-route tool can search a huge implementation space. A signoff tool can tell a team whether a candidate meets a demanding set of physical constraints. But no downstream optimizer can recover an architecture that was never proposed.

Upstream optionality matters because the largest design gains often occur before the implementation problem hardens. A different memory hierarchy, precision choice, accelerator boundary, pipeline strategy, or HW/SW partition can dominate a local downstream optimization. Better downstream tools can't compensate for choices that were excluded from the search space.

AI makes upstream exploration more valuable because proposing alternatives is becoming cheaper. An agent can rewrite, compile, simulate, synthesize, compare, reject,

and retry continuously. It can operate representations that humans may find tedious. It can keep track of previous experiments. It can ask downstream tools for evidence earlier.

Shift-left methods extend that logic by moving downstream information and correlated proxies earlier in the flow while making fidelity itself a design constraint. [173]

At the same time, AI does much less to reduce the difficulty of reproducing the strongest downstream assets. Process models, foundry relationships, qualified flows, accumulated corner cases, integration, support, and trust are specialized assets that generic models don't supply. They're long-lived technical capital.

In the emerging architecture, some intelligence about what to try next can move above mature EDA engines without making those engines dispensable. Synopsys, Cadence, Siemens, OpenROAD-based research, CEA architecture exploration, and multiple university systems all fit versions of this pattern. [1, 2, 15, 182]

Physical design is changing too. Learned placement, differentiable optimization, early physical prediction, and AI-assisted analog design all show that downstream domains can become more programmable. But a clever model may predict signoff. It doesn't automatically become signoff.

Recent reviews map both human-guided AI across the analog flow and the open deterministic layout infrastructure that makes layout-in-the-loop optimization possible. [170, 174, 175]

Analog also resists a simple front-end/back-end split. Expert reasoning and high-fidelity physical evaluation interact throughout the design process, and accurate circuit simulation can be expensive even early. What matters more is how reversible a decision is, how authoritative the available evidence is, and how much that evidence costs—not whether the work sits in a nominal front end or back end.

1.2 A New Computational Actor Enters the Design Flow

The new computational actor is an agentic system that works across a toolchain as a persistent experimenter: proposing work, invoking tools, interpreting evidence, retaining state, and choosing what to try next.

A useful abstraction is a five-stage loop:

Stage	What happens
Propose	Propose architecture choices, RTL edits, constraints, scripts, parameters, verification collateral, or flow changes.
Execute	Established tools execute those proposals.

Judge	Simulation, formal, synthesis, P&R, DRC, timing, and other evaluators return evidence.
Remember	The system records what happened.
Adapt	A repair or search policy changes, and the next experiment is chosen.

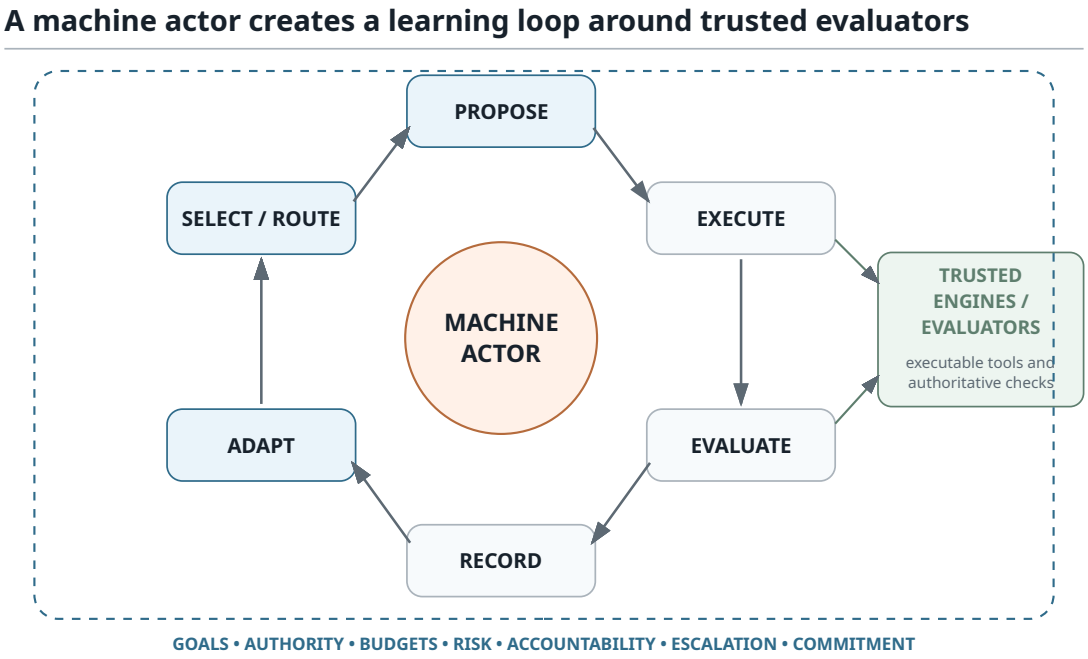


Figure 2. The new actor isn't automatically the judge. It proposes and adapts around executable tools and evaluators while humans still set goals, govern risk and budgets, decide when to escalate, and retain the right to commit.

The propose–execute–judge–remember–adapt loop is increasingly visible in real systems.

Recent surveys organize agentic EDA around handoffs among specialized producers and consumers, hierarchical feedback, persistent context, and acceptance evidence; an NSF workshop report separately identifies data access, benchmarks, integration, and expert knowledge as shared infrastructure needs. Together, the surveys and workshop report map the field and its open problems. [95, 96, 97]

Appendix G provides a multilingual guide to surveys and reviews covering LLM and agentic EDA, shift-left methods, intelligent routing, FPGA automation, high-level synthesis, and analog design.

Earlier systems such as ChatEDA and ChipNeMo show the lineage from tool-using agents and domain-adapted engineering assistance to today's persistent orchestration. [111, 112]

Cadence ChipStack orchestrates multiple engineering tasks over native design and verification tools and has entered named enterprise deployment. [3] Siemens describes long-running agents that continuously validate their actions against physics-based EDA tools rather than relying on model confidence alone. [15] Synopsys's public progression runs from DSO.ai optimization and search, through Synopsys.ai Copilot assistance and AgentEngineer workflows, to Synopsys Autopilot and the announced GPT-Synopsys specialized model. Synopsys and OpenAI describe GPT-Synopsys as operating Synopsys tools, interpreting results, implementing changes, and iterating toward verified outcomes for engineer review. [1, 69, 70, 72, 92, 94] Rule2DRC uses an LLM to generate DRC scripts but judges them by executing them against real layouts. [181]

Verification is a natural early beachhead because the evaluator is unusually machine-readable. Regression results, logs, waveforms, assertions, coverage, formal counterexamples, and pass/fail tests already exist in executable form. A machine can often make progress without being allowed to commit a physical design.

The same propose–evaluate–remember loop applies more broadly. A hardware generator becomes an action space. Synthesis becomes an evaluator. Place-and-route becomes a source of physical evidence. SPICE becomes an expensive circuit-level evaluator. A design database becomes persistent state. A skill library becomes memory.

The greater the consequences of the model's freedom, the more important the evaluator becomes.

Recent work makes the evaluator boundary more explicit:

Benchmark	Evaluator boundary
Long-Horizon Analog Design Bench	Uses an isolated hidden verifier after two-hour schematic-design attempts.
BEHAVE	Checks both agent-written RTL and behavioral models against hidden references.
PolyHDL	Uses equivalence-based oracles to expose synthesis and simulation defects.
SpecRead	Separates specification comprehension from code generation.

These evaluators answer different questions, and none by itself establishes complete signoff or silicon readiness. [116, 117, 141, 142]

Formal verification makes the separation between evaluator quality and generation or repair quality especially clear. A solver can establish that a candidate violates a property and can often produce a counterexample. But Artificial Incorrectness shows that having a strong judge doesn't automatically give the LLM a strong repair policy. [8] Generator quality, evaluator quality, and repair policy are separate technical assets.

An open-silicon verify-before-ship architecture provides a narrower example: deterministic engines remain the core, an agent operates the flow, and a gate preserves

verified results before release. Its two public tasks document the architecture at narrow scale, below production-scale autonomous signoff. [103]

WaveletECO and ResizerAgent show controllers choosing repair or timing strategies around fixed EDA engines. Both are evidence of bounded optimization loops rather than complete signoff. [143, 144]

As autonomy increases, human governance doesn't disappear from the architecture. Current production-oriented systems still rely on explicit goals, budgets, guardrails, escalation paths, accountability, and people with the right to commit. Current public evidence and established human-automation research support an architecture in which agents receive bounded authority and people retain escalation and commitment rights. [3, 15, 68] Control of the loop can be split among different actors.

Actor	Common control or contribution
Incumbent provider	Tool access, product knowledge, evaluator integration, and the design-organization interface.
Design organization	Intent, private design history, accumulated methodology and domain experience, internal tools, and cross-vendor workflow state. [70, 71]
Independent provider	Neutral orchestration, domain expertise, agent infrastructure, or a specialized representation.
Hybrid	Organization-specific engineering context and internal infrastructure coupled to incumbent agent/tool machinery; NVIDIA and Cadence provide one current example.

A persistent computational experimenter can operate around the engines instead of replacing them. Ownership of the surrounding loop is therefore separate from ownership of the engines.

1.3 Evaluator Economics: What Evidence Should the System Buy Next?

Once proposal and search become cheap, the next scarce decision is often which evidence to buy: which evaluator to invoke, when stronger evidence is worth its cost, and when to stop or escalate.

Established disciplines already study versions of this allocation problem. The following table separates their native questions, contributions, and the additional problem evaluator economics addresses. [22, 23, 24, 25, 51, 67, 188, 189]

Antecedent	Native question	Contribution	Evaluator economics adds
Value-of-information analysis	Is more information worth its cost?	Expected decision improvement versus acquisition cost. [22, 188, 189]	Treats information as one form of evaluator evidence.
Engineering design decision analysis	Is another analysis or source worth running?	Cost, quality, and sequential information-source choice. [22, 189]	Generalizes from analyses and sources to evaluator portfolios and escalation.
Multi-fidelity optimization	Which fidelity should be evaluated next?	Cost-aware management of cheap and expensive approximations. [23]	Extends beyond one response at different fidelities; evaluators may answer different questions.
Value-of-computation analysis	Is another computation worth the resources?	Metareasoning about limited computation. [24]	Connects computation choice to engineering evidence, stopping, and escalation.
V&V / credibility	How credible must evidence be for this decision?	Rigor commensurate with reliance and consequence of error. [25]	Adds decision rights and consequence to evaluator allocation.

A broad multi-fidelity survey reviews adaptation, fusion, and filtering methods while emphasizing that even the highest-fidelity model remains an approximation. Cost-aware Bayesian optimization further shows why dividing expected improvement by cost can overfavor cheap evaluations when valuable optima are expensive. [100, 101]

Value-of-information analysis, engineering design decision analysis, multi-fidelity optimization, value-of-computation analysis, and risk-informed verification and validation motivate evaluator economics: allocating effort among evaluators that answer different questions and differ in cost, latency, coverage, fidelity, authority, and risk. Cheap search makes stopping and escalation policy as important as candidate generation.

Concurrent work calls the same resource problem validation congestion: abundant generated candidates compete for scarce validation capacity, and multi-fidelity rankings determine which candidates receive stronger checks. [118] In analog routing, APEC-Route provides a concrete proxy-escalation example: learned capacitance estimates guide search while field-solver references supply stronger evaluation. [140]

Economics supplies another antecedent: as prediction becomes cheaper, judgment and complementary organizational inputs can become relatively scarcer, with consequences for contracts and firm boundaries. EDA differs: proposal generation is not prediction, and qualified engineering evaluation is not generic judgment. The same relative-scarcity mechanism makes evaluation and organizational judgment more valuable as proposal generation becomes cheaper. [102, 113]

EVALUATOR ECONOMICS

Evaluator economics is the engineering concept of allocating evaluation effort among mechanisms that differ in cost, latency, fidelity, authority, coverage, and risk, including deciding when inexpensive evidence is sufficient and when the expected value of greater certainty justifies escalation to a more costly evaluator.

Practical question: What evidence should the engineering system buy next?

The eval.qa guide uses “evaluator economics” for the staffing, quality, and annotation costs of building AI-evaluation teams. [74] This report uses the term for engineering evidence allocation.

Fidelity and Authority Are Different Dimensions

It’s tempting to picture evidence on one ladder: rough approximations at the bottom and increasingly faithful answers at the top. But fidelity and authority are different properties of evidence. That distinction is general, not peculiar to EDA. [25, 75] EDA makes it unusually visible because design flows contain many different evaluators, qualification boundaries, and explicit signoff decisions.

Fidelity asks how closely an evaluation represents the phenomenon, behavior, or result it’s standing in for. Authority asks what engineering question the methodology accepts that result as settling. A result can have high authority for a bounded question without being a detailed physical model, and a high-fidelity analysis can reveal a great deal without settling the engineering gate.

Qualified signoff DRC makes the distinction concrete. A foundry-qualified DRC run can authoritatively answer whether a layout satisfies the encoded design rules for that signoff flow. Its authority comes from the qualified rule deck and methodology. It isn’t trying to reproduce lithography, etching, deposition, process variation, and the rest of fabrication physics in detail. [48]

The opposite case is just as important. A detailed process, lithography, or TCAD analysis can model particular physical mechanisms much more directly. Engineers may use that kind of analysis to understand why a violation occurs, how a structure is likely to behave, or which physical mechanism is driving a problem. That richer physical picture can be exactly what the team needs, even when the analysis is expensive. But unless that

analysis is part of the qualified signoff methodology for the question at hand, physical fidelity by itself doesn't give the result signoff authority. [49, 212]

The distinction becomes clearest in a failure loop. An authoritative evaluator says that the design can't proceed and reports a violation. Engineers then spend more resources on a higher-fidelity investigation because they need to understand why. The higher-fidelity analysis helps diagnose the mechanism and guides the repair. The repaired design then returns to the authoritative evaluator for closure. The extra fidelity was purchased for understanding, not for permission.

Evidence can settle an engineering question without giving anyone permission to act. Organizational authority determines who may approve a tapeout, accept a risk, spend a budget, or commit a consequential change. [76] Those decisions belong to the governance of the engineering system, not to the strength or fidelity of the evidence.

The distinction between fidelity and authority changes how an engineering system allocates evaluation effort. The system isn't simply climbing from cheap, low-fidelity evidence toward expensive, high-fidelity evidence. Sometimes it spends more to obtain a better physical explanation. Sometimes it spends to obtain evidence with enough authority to close a particular engineering question. Sometimes those needs coincide, and sometimes they're served by different evaluators. The routing problem is therefore not "which evaluator is highest on the ladder?" It is "what evidence does this decision need next?"

Recent hardware-modeling work reinforces the boundary: an abstraction can answer its intended question efficiently only when its assumptions and validity range remain explicit. [133]

Fidelity and Authority Answer Different Questions

Example	What it gives you	What it doesn't give you
Qualified signoff DRC	Authority to settle encoded design-rule compliance	A detailed physical model of fabrication
Detailed physical/process analysis	High physical-fidelity evidence for diagnosis and understanding	Signoff authority merely because the analysis is physically detailed

Evaluator selection is iterative:

Condition	Action
Inexpensive evidence is sufficient for the decision	Use inexpensive evidence.
Available evidence is sufficient	Stop.

Uncertainty or consequence makes stronger evidence worth the cost	Escalate to stronger evidence.
A decision has been made	Preserve what was learned so the next decision begins better informed.

Evaluator economics is a routing and stopping problem as much as an accuracy problem. At sufficiently high throughput, scarcity can move again toward commitment.

Chip design makes evaluator economics unusually visible. Its evaluators span enormous differences in cost and latency, and they differ in their fidelity and in what they're authoritative about. Compilation can reject invalid artifacts; formal can settle specified properties; DRC can settle encoded geometric rules; signoff and SPICE answer narrower physical questions at much higher cost; silicon supplies empirical evidence at still another level.

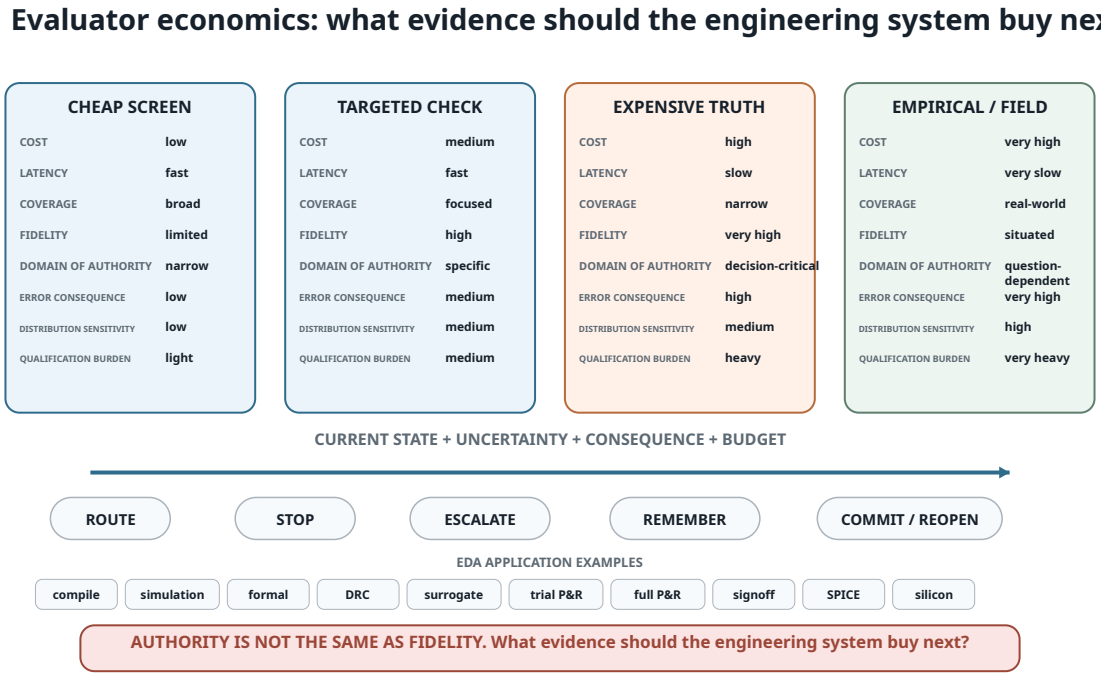


Figure 3. Evaluator economics treats evaluation as a resource-allocation problem: which evidence to obtain next, when stronger evidence is worth its cost, when available evidence is sufficient, and when to stop or escalate. Different evaluators can be authoritative over different questions, so authority can't be reduced to a single fidelity ladder.

Evaluators combine cost, fidelity, and authority differently. Some are authoritative within a defined scope; others provide useful evidence without settling the decision:

Evaluator	**What it can settle / enable**	**Cost, authority, and main limitation**
Syntax and compilation	Reject invalid artifacts quickly.	Very low cost and latency; little evidence about intended behavior.

Evaluator	What it can settle / enable	Cost, authority, and main limitation
Simulation and tests	Provide machine-readable behavioral feedback over exercised workloads and tests.	Low to medium cost; rich but incomplete state-space coverage.
Formal and DRC	Formal can establish specified properties; DRC can establish encoded geometric rules. Rule2DRC shows the value of execution-guided selection. [181]	Authority is high within the specified rule/property scope, but strong judges don't automatically provide strong repair policies; Artificial Incorrectness makes that gap explicit. [8]
HLS synthesis	Provides a deterministic compiler/synthesis loop and structured inner-loop evidence.	Medium cost; strong for synthesizability, but historical evidence shows a persistent QoR gap versus strong RTL. [9]
Learned surrogates and virtual prototypes	Rank or evaluate many candidates cheaply; VPSim reports up to 600 MIPS for architecture exploration. [182]	Fidelity is model-dependent, and authority is limited to the questions for which the model has been validated. Analog multi-fidelity work reports replacing ~96.5% of candidate SPICE evaluations and ~20× lower simulation time while retaining selective SPICE checks. [19]
Trial physical implementation	Provides stronger physical evidence than analytical or learned estimates.	Medium cost; more accessible and programmable than full advanced-node production flow, but less authoritative than production signoff.
Full P&R, signoff, high-fidelity SPICE	Answer demanding physical, timing, and circuit questions under qualified assumptions.	High cost and latency; substantially more authoritative for the questions they cover.
Silicon	Provides ultimate empirical behavior.	Extreme cost and latency; unsuitable for ordinary inner-loop search.

Differences in evaluator cost and fidelity lead naturally to multi-fidelity search.

CASE STUDY 1 — COST OF STRONGER EVIDENCE: Multi-fidelity analog optimization

Dimension	Finding
Technical mechanism	A learned surrogate screens or ranks many candidate circuits. Selected candidates are sent to SPICE for higher-fidelity evaluation, and those results can update the surrogate and the search policy.
Evaluator and evidence structure	The surrogate provides fast, approximate predictions. SPICE provides stronger circuit-level evidence under the device models, solver settings,

Dimension	Finding
	and analyses being used; it's still a model of circuit behavior, not silicon.
Ownership and control paths	The cited studies don't establish a dominant ownership pattern. The optimization layer could be built inside a design organization or supplied externally, while the higher-fidelity simulator can remain incumbent or internal infrastructure.
Evidence	One 2026 study evaluates four operational-amplifier topologies. Its surrogate-guided mode replaces about 96.5% of SPICE calls and reports about 20x lower total simulation time; its adaptive multi-fidelity mode reduces SPICE use by roughly 90%. [19] A separate 2024 study on three analog circuits reports 56%, 59%, and 83% reductions in SPICE calls. [185] Other studies report more than 15x speedup for a Bayesian Monte Carlo surrogate and 3.49–8.18x speedup from asynchronous batch optimization. [186, 187] Those results measure different mechanisms and shouldn't be treated as the same result.
Demonstrated engineering value	The demonstrated value is engineering leverage: fewer high-cost simulations and shorter optimization runs in the tested research settings. That can support more search for the same simulation budget, but the cited studies don't measure generalized production dollar savings or provider economics.
Potential durable advantage	Reusable surrogates, training data, calibration history, and routing policies can become engineering assets, but the cited work doesn't show that they create a durable advantage across projects or organizations. Changes in topology, process node, objectives, or simulator settings can require retraining or requalification. Control could remain inside a design organization, sit with an incumbent platform, or be split between an external optimization layer and an authoritative simulator.
What remains open	Production-scale robustness; deployment and maintenance burden; retraining or requalification cost as designs, nodes, tools, and objectives change; transfer across projects or organizations; and whether routing knowledge creates switching cost or durable external pricing power.

Operationally, the evaluator hierarchy in Figure 3 becomes a simple loop:

Stage	Action
Generate	Generate many candidates.
Filter	Use cheap evaluators to reject obvious failures.
Escalate	Escalate uncertain or promising candidates.
Verify	Invoke higher-cost evidence selectively.
Learn	Feed that evidence back into the search policy.

A multi-fidelity evaluation system can create value at several leverage points:

Leverage point	Source of value
Generator	Proposes better candidates and expands the useful search space.
Evaluator	Makes evidence cheaper or more available, allowing more candidates to be judged.
Escalation policy	Concentrates higher-cost evidence on uncertain or promising candidates when stronger evidence is worth buying.
Repair policy	Turns evaluator feedback into a better next candidate rather than merely detecting failure.
Engineering memory and routing	Preserves results, learns which evaluator is needed, and lets the next decision begin better informed.

Differences in evaluator cost, fidelity, and authority help explain why domains reach autonomy at different rates. Verification has relatively cheap strong judges. Analog has expensive but usable simulators plus improving surrogates. Architecture has fast models but lower fidelity. Signoff has high authority and high cost. The categories differ less by how much AI they contain than by how cheaply a machine can obtain evidence strong enough to keep moving.

Evaluator cost and authority shape how much search is practical, where autonomy arrives first, and which evaluators control strategically important evidence. Public evidence doesn't yet show that evaluator routing by itself creates a durable standalone business advantage.

Differences in evaluator cost and authority raise a historical question: which automation approaches were held back mainly by constraints AI can now relax, and which remain limited by constraints AI leaves largely intact?

1.4 Which Constraints Does AI Actually Relax?

Many current AI-driven EDA approaches have clear predecessors. What matters is which constraints limited those earlier approaches and whether AI relaxes those constraints.

Architecture	Original objective	Prior limiting constraints
High-level synthesis	Move design above RTL and let a compiler produce implementation.	QoR gap, hardware-aware restructuring, predictability and debug, directive expertise, and workflow adoption.
Hardware construction languages and generators	Represent families of circuits and reusable structural intent rather than isolated instances.	Language and ecosystem learning cost, generator debugging, methodology change, and entrenched RTL workflows.
Virtual prototyping	Move software and architectural feedback earlier through executable system models.	Model-building and maintenance cost, calibration, specialist operation, and fidelity

Architecture	Original objective	Prior limiting constraints
		limits.
Design-space exploration	Search alternatives systematically rather than rely only on human intuition.	Workflow integration, usability, solver and simulation cost, organizational adoption, and commercialization difficulty.
Analog automation	Encode and reuse sizing rules, topology knowledge, generators, and expert procedures.	Tacit knowledge, process dependence, expensive simulation, and limited transfer across topologies and nodes.
Formal methods	Turn correctness into executable evidence through proofs and solvers.	Specification burden, solver scalability, and expert interpretation of counterexamples.
Unified design databases	Preserve design state across fragmented tools.	Cross-tool integration and organizational fit.

These ideas weren't waiting for AI because everyone had missed an obvious insight. They faced real constraints. Some were expensive for humans to operate. Some imposed training and workflow costs. Some faced QoR or fidelity limits. Some had hard solver-scaling limits. Some were process-dependent. Some never fit organizational practice.

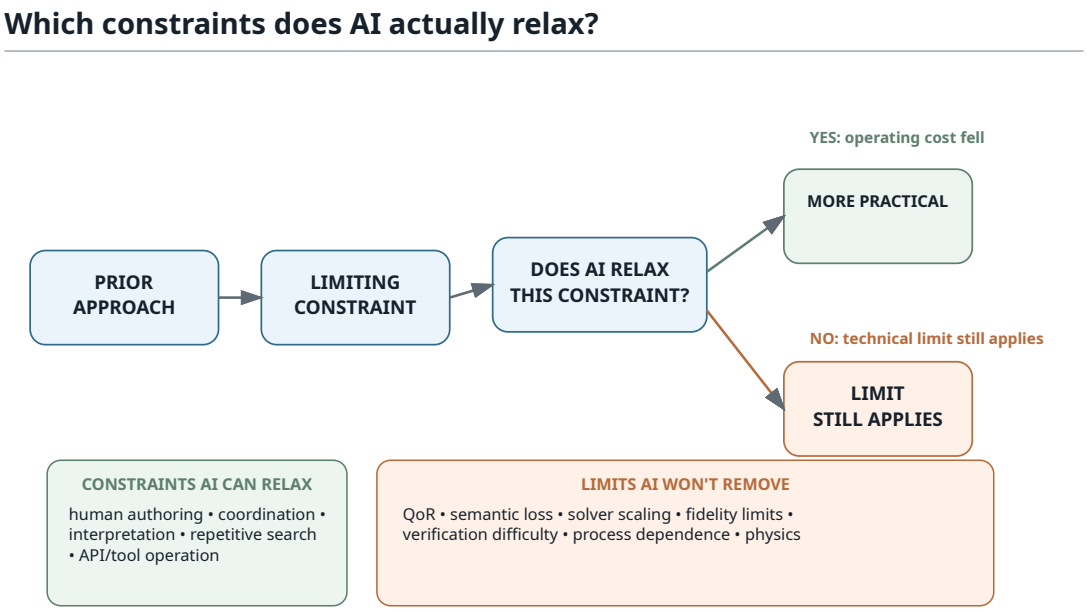


Figure 4. AI can relax some of the constraints that limited an EDA approach. If the limiting factor was expert human effort, then the economics can change substantially. If the limiting factor was QoR, solver complexity, fidelity, process dependence, or physics, then the economics don't change much.

Architecture	What AI relaxes	What still limits the approach	Evidence / implication
High-level synthesis	Restructures high-level code, interprets reports, explores directives, and runs more experiments around a deterministic compiler/synthesis loop.	Algorithms or mappings poorly suited to HLS don't become suitable; downstream physical constraints still apply; QoR can still lag expert RTL.	A 2018 survey across 46 papers and 118 applications found development time around one-third of RTL and average productivity above four times RTL, while RTL generally retained better QoR. [9] Japanese reviews trace the same lineage through the mechanisms that made HLS practical and the languages and systems used to raise FPGA design above RTL. [167, 168] AI can reduce the expert effort needed to operate and explore around the compiler, but it doesn't eliminate the QoR gap.
Hardware generators and construction languages	Creates and modifies generators, searches parameter spaces, and translates intent into structured design actions without requiring the same human investment in learning the language and methodology.	Generator bugs can propagate widely; verification and downstream integration are still mandatory.	Structural reuse and parameterized design become attractive machine action spaces, although production-market evidence remains uneven.
Virtual prototyping	Maintains models, runs workloads, interprets results, and schedules experiments continuously.	Model fidelity and correlation with the eventual implementation still determine whether fast feedback is useful.	CEA's VPSim and related architecture-exploration work demonstrate the use of executable models to explore architectural choices before RTL. [182] AI can reduce the effort required to build and use those models, but they still have to predict the later implementation accurately enough to guide good decisions.
Analog automation	Ingests design knowledge, generates candidates, searches, and orchestrates simulator loops,	High-fidelity simulation, process dependence, topology variation, generalization, and silicon behavior	French CAIRO/OCEANE work and Iskander, Louërat, and Kaiser's dependency-graph approach to operating-

Architecture	What AI relaxes	What still limits the approach	Evidence / implication
	reducing the expert effort needed to encode knowledge and explore alternatives.	continue to impose substantial cost.	point computation and design-plan generation provide older executable-knowledge lineage. A current Japanese program explicitly aims to capture analog intent, constraints, and evaluation criteria for reuse. [110, 114] Reinforcement learning, Bayesian optimization, and surrogate models extend this boundary; physics still sets the limit.
Formal methods	Assists specification, tool use, and counterexample interpretation.	Solver complexity doesn't disappear, nor does the gap between proving a design wrong and finding a good repair.	Artificial Incorrectness shows strong formal validation without reliable repair from counterexamples alone. [8] AI complements formal methods; it doesn't remove their hard problems.

AI can make some of these approaches more practical by reducing the expert effort needed for authoring, coordination, interpretation, and search. But QoR limits, fidelity limits, solver complexity, process dependence, and physics will still put real limits on how far they can go.

CASE STUDY 2 — HIGHER-LEVEL SEARCH: SoC Planner and agentic HLS

Dimension	Finding
Technical mechanism	SoC Planner is an integrated flow: CEA A-DECA explores and evaluates architectures, Innova PDM coordinates workflow and engineering data, and Defacto SoC Compiler turns selected architectures into synthesizable RTL. UCLA's AHRR is a separate research workflow in which an agent works in HLS C++, Vitis HLS generates RTL, and an agent can refine that RTL. The shared pattern is higher-level search coupled back to deterministic downstream tooling.

Dimension	Finding
Evaluator and evidence structure	SoC Planner evaluates architecture candidates with analytical models and simulation before selected configurations move downstream to RTL. In AHRR, Vitis HLS is primarily a translator; candidate quality is judged by correctness tests, RTL simulation, simulated cycle count, post-route clock period, and resource ceilings on an AMD U55C. These evidence sources differ in fidelity and authority, and the FPGA results aren't ASIC production signoff.
Ownership and control paths	SoC Planner spans CEA, Defacto, and Innova; Defacto entered a Siemens ownership transition in 2026; Rise demonstrates an independent-vendor route. Internal generators, open compiler infrastructure, public/research systems, independent vendors, and incumbent platforms are all live paths. Public evidence doesn't establish a dominant one.
Evidence	SoC Planner first-party sources describe an integrated architecture-to-RTL flow, two industrial validation scenarios, and up to 30–40% shorter exploration cycles. [13] Ding et al. report, on an 11-task FPGA benchmark, 2.62× geometric-mean execution-time speedup for AHRR over Direct RTL on valid paired cases and 2.31× for Agent-HLS alone. [50] A historical HLS survey reports average development time about one third of RTL and more than 4× designer productivity, while RTL QoR remained better on average. [9]
Demonstrated engineering value	The demonstrated value appears in three different metric families: SoC Planner reports shorter exploration cycles in early-adopter validation; UCLA AHRR reports faster generated-hardware execution in a controlled FPGA benchmark; historical HLS literature reports human development-time and productivity gains. These results show engineering leverage at higher abstraction, but they don't establish generalized dollar savings, reduced human burden in the UCLA experiment, or a standalone software revenue pool.
Potential durable advantage	Reusable representation/compiler infrastructure, workflow integration, architecture/evaluation data, and generation/refinement patterns could become durable engineering assets. Public evidence doesn't yet show cross-project compounding, switching costs, or durable pricing power, and changes in target, toolchain, design class, or QoR requirements can limit transfer.
What remains open	Broad production adoption; repeatability and generalization of agentic results; engineering effort to build and maintain these flows; QoR and maintainability relative to strong RTL; transfer across designs, devices, and toolchains; who captures value; and whether design organizations prefer external higher-level providers or internal generator/HLS infrastructure.

2. What Exists Now

2.1 Where Autonomy Arrives First

Autonomy doesn't arrive on a single maturity curve. It arrives first where a machine can obtain useful evidence cheaply, execute the relevant checks, and trust the results enough to keep moving.

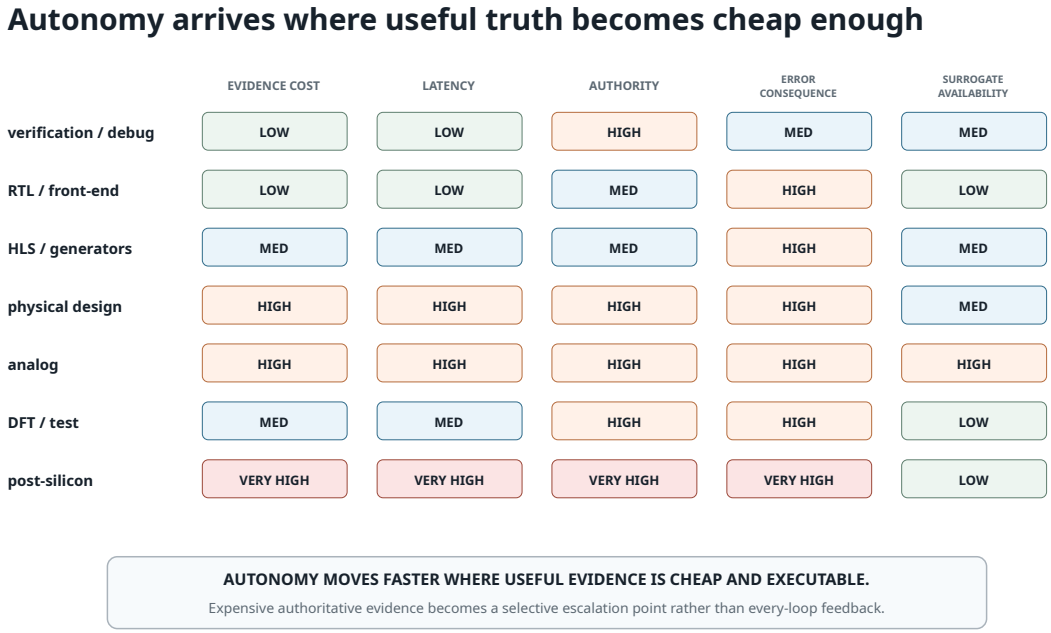


Figure 5. Model capability alone doesn't determine autonomy. A domain becomes easier to automate when useful evidence is cheap, executable, and authoritative enough for the decision being made.

Domain	Evaluator structure	Why autonomy is easier or harder	Limit AI doesn't remove
Verification and debug	Regression results, logs, waveforms, assertions, coverage, tests, and formal counterexamples provide executable feedback.	Relatively strong, machine-readable judges let a system run, compare, reject, and retry without committing a physical design.	Evidence is bounded by the tests, properties, workloads, and specifications being exercised.
Front-end RTL generation	Compilation and simulation provide fast feedback, but they provide much less complete evidence of intended behavior across all relevant conditions.	Candidate generation is cheap and executable; establishing semantic correctness is substantially harder.	Hardware-specific failures persist in sequencing, handshakes, assignment semantics, and state modeling. [17]
HLS and generators	Compilers enforce representation semantics and produce deterministic implementation artifacts that can be evaluated downstream.	A structured action space and deterministic compilation make repeated agent operation attractive.	QoR and maintainability still have to compete with strong RTL methodology.

Domain	Evaluator structure	Why autonomy is easier or harder	Limit AI doesn't remove
Physical design	Placement, routing, sizing, and physical-aware optimization expose objective functions and executable tools, with downstream validation available.	The domain is highly programmable and has measurable objectives.	Generalization across designs and process nodes is difficult; production use still requires strong downstream validation.
Analog	High-fidelity simulation supplies strong but expensive evidence; learned surrogates and Bayesian methods provide cheaper approximate evaluations.	Multi-fidelity routing can reserve expensive simulation for uncertain or promising candidates and greatly expand search.	Process dependence, topology variation, high-fidelity simulation cost, and silicon behavior continue to limit what's practical.
DFT, test, and post-silicon	DFT and test provide structured generation and checking; post-silicon work provides rich logs and observable failures.	Repetitive, correctness-sensitive work with explicit checks is comparatively automation-friendly.	Post-silicon evidence comes with expensive physical context, and generated test structures still require trustworthy checking.

Across these domains, autonomy turns on a practical question: how cheaply can the machine know whether it's right? Evaluator economics begins there. Differences in evaluator cost and authority determine how much search a system can afford, which forms of truth become relatively scarce, and where control and value can accumulate.

2.2 AI-driven EDA Is Already Several Markets

Public evidence already spans several distinct AI-driven EDA markets.

AI-driven EDA market categories have different evaluator structures: different costs of truth, domains of authority, executability, privacy constraints, integration burdens, and ownership tendencies. That gives verification agents, analog optimization, higher-level design, physical optimization, workflow state, and incumbent platforms different economic shapes.

The examples show where technical architectures are being tried and commercialized while long-run winners and market structure remain open.



Figure 6. AI-driven EDA already spans several markets. Different wedges act on different representations, evaluators, buyers, control points, compounding assets, and pressures toward platform integration or internalization by design organizations. Examples in the open/shared category include OpenROAD/CIRCT/XLS and iEDA/AiEDA.

Market wedge	Architecture / evaluator structure	Economic boundary
Incumbent agent platforms	Cadence, Siemens, and Synopsys extend established engines into assistants, orchestration, goal-directed engineering, and specialized model layers. [3, 15, 69, 70, 72, 92, 94]	Native engine access, documentation, support history, customer relationships, and qualified flows are advantages; design environments vary widely and are strategically sensitive, which limits complete control.
Front-end design agents	Systems such as ChipAgents and academic spec-to-implementation agents work across specification, RTL, verification collateral, microarchitecture, and debug. ChipAgents reports named production workflows, on-premises deployment, and customer-controlled design data. [145, 146, 148, 150, 153]	High iteration before physical signoff is attractive. Named results establish bounded workflow gains; maintainability, integration burden, renewal, and differentiation require separate evidence.

Market wedge	Architecture / evaluator structure	Economic boundary
Verification and debug	Bronco, MooresLabAI, ChipAgents verification workflows, Rule2DRC, and related systems work against logs, coverage, assertions, regressions, formal engines, waveforms, and failure artifacts. ChipAgents reports bounded verification and root-cause workflow results. [147, 151, 152]	Strong executable evidence makes verification an early autonomy beachhead; production value still depends on integration, coverage, and human review.
Higher-level design	Rise, HLS-based systems, XLS, Chisel, ITDA-style structured generation, and related approaches change the representation rather than merely improving direct RTL generation.	Larger action spaces compete with adoption friction, QoR risk, and RTL-methodology integration.
Physical optimization	RL, Bayesian optimization, graph methods, differentiable optimization, and learned predictors optimize placement, routing, sizing, and physical-aware decisions.	Technical value is real; generalization, downstream correlation, and incumbent response still shape the outcome.
Analog automation	RL, Bayesian optimization, generators, and surrogates reduce search cost while high-fidelity simulation often supplies the evaluator.	Autonomy is still limited because methods may not generalize across processes, topologies, or operating conditions, and high-fidelity simulation remains expensive.
Workflow and system of record	DSM Pro, Defacto-style persistent design databases, Innova, and incumbent data layers focus on authoritative state, provenance, configuration, resource use, and history.	The visible agent may become easier to replicate than the underlying data and infrastructure that maintain authoritative design state.
Surrogates and models	Learned models can provide cheaper approximations of expensive physical or circuit evaluations, allowing larger search loops to use high-cost analysis more selectively.	They may change exploration economics without owning the user interface.
DFT, test, and post-silicon	Automation extends into test, validation, bring-up, and diagnosis.	Evaluator, buyer, and integration requirements differ materially from front-end design.

Market wedge	Architecture / evaluator structure	Economic boundary
Open infrastructure	OpenROAD, CIRCT, XLS, Chisel, Amaranth, OpenFASoC, Hammer, iEDA/AiEDA, and related projects expose programmable infrastructure and lower experimentation cost. [16, 36, 38, 39, 40, 42, 156, 193, 194, 195]	They can make inner loops cheaper without themselves becoming standalone vendor categories.

What separates these categories economically is where they create leverage: some reduce the cost of evaluation, some operate trusted engineering engines, and some maintain the state and infrastructure around them. Commercial evidence remains uneven across these categories.

AI-driven implementation optimization is clearly production-grade. Synopsys reported its first 100 commercial DSO.ai tape-outs by February 2023 and DSO.ai in production in about 20% of its top 100 Place & Route customers as of the end of 2023. [1, 179] Cadence Cerebrus has named production customers and more than 1,000 tapeouts. [2, 3]

Agentic verification has also crossed the threshold from demo to enterprise deployment. Cadence has reported more than ten early ChipStack customers, including Altera, NVIDIA, Qualcomm, and Tenstorrent. [3] ChipAgents reports named workflows at Andes, eMemory, Whalechip, and Ambiq, bounded gains in elapsed time, engineering effort, and test generation, expansion beyond evaluation at Ambiq, and more than 200 semiconductor-enterprise deployments as of October 2026. [6, 145, 146, 147, 148, 149]

Public commercial evidence is thinner in other categories. Higher-level representations, neutral orchestration, persistent engineering memory, and analog automation have credible technical rationales, but public evidence on renewal, switching costs, and durable pricing power remains limited.

A technically important capability may not support a standalone company. The company-shaped/capability-shaped distinction describes current commercial forces, not a permanent end state.

Commercial category	Defining condition	Likely outcome
Company-shaped	A focused product can attract repeated purchases, expand within organizations, create switching costs, and build an asset that grows more valuable with use.	A durable independent company is economically plausible.
Capability-shaped	A capability can be technically important even when it doesn’t support a standalone product. Its value may instead be realized when an incumbent integrates it	The capability may remain valuable without developing into a durable standalone software business.

Commercial category	Defining condition	Likely outcome
	into a broader platform, a design organization uses it internally, an open ecosystem maintains it as shared infrastructure, or a provider delivers it through services.	

Five commercial questions help distinguish the two:

Commercial criterion	Question
Repeated purchase	Does a buyer repeatedly purchase it rather than build it?
Vendor asset that grows with use	Does repeated use build knowledge, data, or other assets that grow more valuable over time? Can the vendor retain them under the relevant IP, privacy, and contractual rules?
Integration depth	Does deeper integration make the product more valuable and harder to replace, or make the capability easier to absorb into a broader platform?
Market durability	Can the product sustain a distinct market if incumbents integrate similar capability into broader platforms or design organizations build it internally?
Customer budget	Do customers have a recurring budget large enough to support a standalone business?

CASE STUDY 3 — INCUMBENT AI OPTIMIZATION IN PRODUCTION: Synopsys DSO.ai

Dimension	Finding
Technical mechanism	DSO.ai uses reinforcement learning to search large digital-implementation spaces. It analyzes data from Synopsys design tools and adjusts design choices, technology parameters, and workflow settings toward multidimensional power, performance, and area objectives. [1, 87, 88]
Evaluator and evidence structure	The search is evaluated through results generated inside Synopsys's digital-implementation flow. Fusion Compiler combines implementation with built-in signoff timing, parasitic extraction, and power analysis, so DSO.ai can act on strong engineering evidence while the underlying implementation and signoff engines remain the source of that evidence. [89, 90]
Ownership and control	DSO.ai and the implementation/signoff engines are Synopsys offerings. The design organization still sets objectives, constraints, workflow policy,

Dimension	Finding
	compute budgets, and approval boundaries around the platform, so product ownership and system-level control can remain distinct. [87, 88, 89, 90]
Emerging specialized-model layer	GPT-Synopsys adds a jointly developed, OpenAI-hosted specialized model above Synopsys tools and evaluators. The announced architecture distributes control across layers: OpenAI hosts the model; Synopsys supplies tools, engineering context, domain agents, and ground-truth engines; customers may use their own agent harnesses; engineers review outcomes. Early technology engagements are underway, but the announcement reports no realized productivity or broad deployment results. [92, 93]
Evidence	Synopsys reported the first 100 commercial DSO.ai tape-outs by February 2023, including named users STMicroelectronics and SK hynix. At its 2024 Investor Day, Synopsys reported DSO.ai in production in about 20% of its top 100 Place & Route customers as of the end of 2023. [1, 179]
Demonstrated engineering value and commercial evidence	Named customer results include STMicroelectronics reporting more than 3x PPA-exploration productivity and SK hynix reporting a 15% cell-area reduction and 5% die shrink. Synopsys separately reported about 20% uplift versus baseline Fusion contract value, with the baseline defined as Fusion contract value before DSO.ai. Together, these provide direct evidence of engineering gains for named customers and commercial value for Synopsys in the reported deployments. [1, 179]
Potential durable advantage	DSO.ai benefits from close integration with Synopsys implementation/signoff infrastructure and from reusable learned state. Synopsys states that learned models can be applied across projects. In a January 2024 related-node retargeting example, reuse of a 5 nm training database at 4 nm reduced the Fusion Compiler workload from 30 parallel jobs over three iterations to 15 jobs in one iteration—a sixfold reduction versus cold start. This bounded reuse shows how learned state can make the optimization capability more valuable with repeated use inside the broader platform. [87, 88, 89, 90, 91]
What remains open	Public evidence establishes production use, named-customer engineering gains, commercial adoption, and bounded reuse of learned state. The remaining questions are how broadly that learned state transfers beyond related designs and how much of the reported contract-value uplift should be attributed specifically to DSO.ai versus the broader Fusion relationship. [1, 91, 179]

3. Control, State, and Technical Capital

3.1 How Is Control Distributed?

The central strategic question is no longer whether agents can operate EDA tools. Public evidence increasingly shows that they can. Control can be split among different actors. The harder question is how that distribution changes as capabilities and constraints change, which control points become economically important, and where durable value begins to accumulate.

Control can be distributed among multiple actors. Decision rights locate authority, resource dependence shapes bargaining power, transaction costs influence organizational boundaries, and complementary assets shape value capture. [65, 76, 77, 78] Different actors can set goals, define proposal policy, grant evaluator access, maintain authoritative design state and engineering memory, choose routing and escalation policy, allocate compute and license budgets, and retain the right to commit.

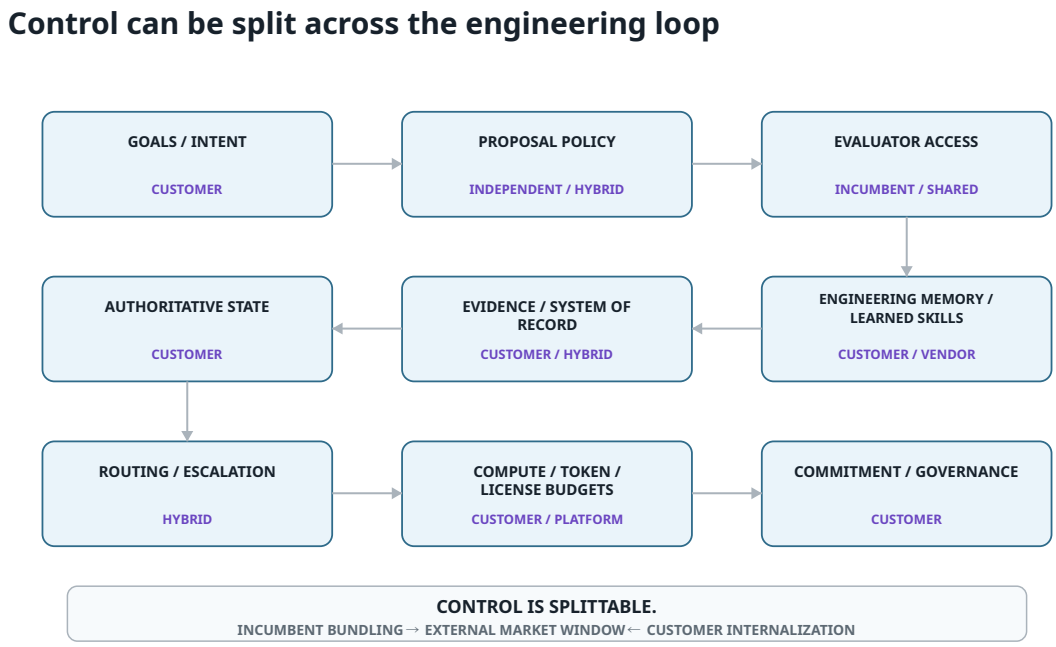


Figure 7. Control of the agent–evaluator engineering loop can be distributed among multiple actors. Incumbent providers, design organizations, independent providers, and shared infrastructure can control different parts of the same engineering system. Independent markets persist when a capability is valuable enough to buy, difficult for design organizations to internalize, and uneconomic for incumbents to incorporate into broader platforms.

Control Points Across the Engineering Loop

Control can be distributed across several parts of the loop:

Control point	Governs
Goals and intent	What the system is trying to achieve and which tradeoffs matter.
Proposal policy and action space	What changes may be proposed and which alternatives are considered.
Tool and evaluator access	Which engines and evidence sources may be invoked.
Authoritative design state	Which design version, constraints, and state count as current.

Evidence and system of record	Which results are retained as authoritative evidence.
Engineering memory and learned skills	What experience persists and can shape later work.
Evaluator-routing and escalation policy	Which evaluator is used when and when stronger evidence is required.
Compute, token, and license budgets	How much compute, model, and tool capacity the system may spend.
Commitment and governance	Who may approve, commit, or govern consequential changes.

An actor can gain durable leverage by controlling several control points even when no party controls the whole loop.

Control plane	Current advantage	Implication for control
Incumbent provider	Major engines, product documentation, support history, customer relationships, integration knowledge, and direct reach into simulation, formal, synthesis, P&R, and signoff.	Natural-language invocation alone may be difficult to sustain as a distinct external product when tool vendors can incorporate comparable interaction patterns into broader platforms.
Design organization	Can build and retain proprietary methodology, design history, internal tools, private data, APIs, state, and cross-vendor workflow context. Google and Apple provide public examples. [5, 18, 183, 184]	When a design organization can create large value by building the capability internally, control can stay inside the organization even without an external software market.
Independent provider	Cross-vendor neutrality, specialized evaluator or representation infrastructure, secure deployment, integration expertise, and reusable provider assets. ChipAgents, Bronco, Rise, Chipmind, and others show the layer is real; the ChipAgents case adds named workflow gains, on-premises deployment, and customer-controlled design data. [6, 26, 27, 29, 145, 146, 147, 148, 150, 153]	An independent provider can retain control when customers value capabilities incumbents don’t readily absorb and design organizations don’t prefer to build. The durable position depends on portable integrations and learning, manageable deployment effort, and customer renewal.

Control plane	Current advantage	Implication for control
Hybrid	A design organization can retain its own context, models, engineering memory, or infrastructure while relying on provider-owned evaluators and agent execution. NVIDIA + Cadence is a public example. GPT-Synopsys shows another hybrid pattern: OpenAI hosts a jointly developed specialized model; Synopsys supplies the tools, engineering context, domain agents, and ground-truth engines; customers may use their own agent harnesses; and engineers review outcomes. [4, 92, 93]	Different actors can control different parts of the loop without any one of them controlling the whole system.

Customer demand, asset retention, security, integration, knowledge transfer, and internal-build economics determine whether a capability supports an independent provider or shifts into a broader platform, the design organization, open infrastructure, services, or sovereign capability.

CASE STUDY 4 — DESIGN-ORGANIZATION CONTROL PLANE: Apple and Google

Analytical dimension	Case finding
Internal integration layer	Google and Apple describe internal CAD and design environments that connect company-specific flows, interfaces, data, metrics, and automation to commercial and internal engineering tools. The internal layer coordinates work across design stages; it doesn’t replace every underlying engine.
Evaluator boundary	Commercial EDA engines and internal analysis or checking tools produce question-specific engineering evidence. The internal environment governs how tools are connected, invoked, monitored, and incorporated into company methodology. Signoff authority still comes from the qualified checks used for the decision.
Organizational control	The design organization controls its internal environment, proprietary interfaces, methodology, flow state, metrics, and decisions about which tools and evidence to use. Dependence remains layered: Google explicitly combines first- and third-party tools, and Apple works directly with EDA vendors. [5, 18, 77, 183]
Google evidence	Google describes its silicon design environment as a coherent system of first- and third-party tools, industry-standard and Google-specific APIs/data formats, scalable flows, shared methodology, and procedural and AI-first approaches. The posting assigns development and deployment responsibility; it doesn’t report realized productivity or ROI. [5]

Analytical dimension	Case finding
Apple evidence	Apple describes signoff dashboards and metrics used across SoC groups, a Signoff AI assistant, custom implementation and signoff tools, ML/GenAI methods, RTL-construction automation, documentation and training, and direct work with EDA vendors. These descriptions show a broad internal CAD capability; they don't measure the resulting productivity or economics. [18, 183, 184]
Economic mechanism	An internal control layer can lower coordination costs, preserve company-specific methodology, and spread tooling across projects and design teams. The sources show internal capability building and stated productivity goals; they don't quantify realized cost, schedule, or ROI. [5, 18, 78, 83, 183, 184]
Durability and dependence	Flows, interfaces, metrics, documentation, and operating knowledge can become more useful through reuse. Their value can also depreciate or require requalification when tools, process nodes, methodologies, or product architectures change. External engines and vendor expertise can remain essential. [5, 18, 60, 61, 62, 63, 183]
What remains open	Public job descriptions don't disclose realized productivity, internal cost, portfolio-wide adoption, procurement rationale, switching costs, requalification burden, ownership of learned AI artifacts, or which functions the companies deliberately buy rather than build.

CASE STUDY 5 — INDEPENDENT AGENT PROVIDER: ChipAgents

Analytical question	What the evidence shows
What does the provider supply?	ChipAgents supplies agent software for specification, RTL, testbench and stimulus generation, simulation orchestration, debug, coverage, and root-cause analysis. The agents operate above customer engineering environments and invoke established EDA tools as evaluators.
What produces the decisive engineering evidence?	Simulators, regressions, coverage, formal methods, timing and design-rule checks, foundry rules, and golden simulations produce the decisive engineering evidence. Agents invoke and interpret these evaluators; engineers set acceptance criteria, review the evidence, and retain signoff authority. [151, 135]
What does the customer control?	Renoir supports on-premises and air-gapped deployment. ChipAgents states that customers control deployment, design data, and security policy and that customer RTL, waveforms, and logs aren't used to train models. These controls establish a data and deployment boundary. Public materials leave telemetry, update procedures, remote support, derived-artifact ownership, and cross-customer reuse to contract and diligence. [150, 153]
What's been demonstrated in named workflows?	Andes Technology — Processor-customization time fell from 8.5 to 6 weeks; BIU verification fell from two to three months to about three weeks; stimulus generation fell from one to two weeks to one to two days across multiple production projects. eMemory — An NVM-IP functional-model workflow used about 80% less effort, fell from four days to five hours, and generated 18% more test patterns. Whalechip — Root-cause-analysis rounds fell from days to 15–60 minutes in a bounded four-bug episode involving 60 hours of

Analytical question	What the evidence shows
	engineering effort. These are participant-reported deployment results for specific workflows, not controlled estimates of typical performance. [146, 147, 148]
What shows adoption and scale?	Ambiq moved from evaluation to deployment across additional engineering teams. As of October 2026, ChipAgents reports more than 200 semiconductor-enterprise deployments. This is a time-stamped provider count, not an audited measure of market share, retention, or deployment depth. [145, 149]
How does the system create customer value?	The reported value comes from reaching credible engineering evidence sooner: generating artifacts, running evaluators, narrowing failure hypotheses, and iterating before another manual cycle. The named reports show bounded gains in elapsed time, engineering effort, and test generation. They provide no category-wide productivity rate.
What does deployment require?	Production deployment still requires integration with customer tools and environments, handling of design-specific corner cases, human guidance, and measurement of return on investment. ChipAgents' own DVCon workshop description identifies each of these as a practical deployment problem, and its RCA account describes remaining limits at subsystem and SoC scale. [151, 152]
What can accumulate?	The provider can accumulate tool integrations, orchestration patterns, evaluator interfaces, deployment methods, and support experience across customers. The customer can accumulate local workflows, acceptance criteria, qualified prompts or procedures, audit trails, and evidence about when the system can be trusted. The reusable value of each asset depends on portability, contractual control, and requalification in the next design context.
Which dependencies and ownership alternatives matter?	ChipAgents depends on customer design state, EDA engines, infrastructure, vendor interfaces, and human decision authority. Customers can internalize orchestration, use bundled capabilities from incumbent EDA vendors, combine providers, or retain a specialist layer while keeping data and execution inside their own environment. The economic position turns on which integrations, learning, and qualified workflows remain portable.
What remains to be established?	Remaining diligence questions include renewal and expansion, independently verified deployment depth, service effort per deployment, telemetry and update arrangements, ownership and reuse of learned artifacts, requalification burden, and customer switching behavior. The published reports establish use and outcomes in specific workflows. These questions determine how much of that use becomes durable provider advantage.

Design organizations may build and retain the following capabilities internally:

Capability	Why organizations may build and retain it internally
Company-specific methodology agents	They encode proprietary methodology, workflow context, and engineering judgment specific to the organization.

Capability	Why organizations may build and retain it internally
Private failure-history databases	Private failure histories can reveal proprietary architecture, recurring failure modes, and how engineers diagnose and fix them. That knowledge can be highly valuable to the organization, and letting it escape can expose both the design and the organization's engineering methods.
Internal evaluator-routing policies	Evaluator-routing policies encode private choices about cost, risk, and methodology, and about when stronger evidence is required. Keeping those choices internal preserves the organization's control over how work is evaluated and escalated.
Proprietary design-state integration	The integration connects directly to internal APIs, data formats, tools, and the design state the organization treats as authoritative. Moving it outside the organization can require exposing or recreating those private interfaces and state.
Cross-project memory coupled to product architecture	Cross-project memory can carry lessons that depend on the organization's product architecture. Moving that memory outside the organization can expose proprietary design knowledge, while the lessons themselves may lose value when separated from the context that made them useful.
Internal search policies	Internal search policies determine which alternatives the organization explores, which constraints it treats as fixed, and where it spends engineering effort. Those choices reflect product and engineering strategy that the organization may not want to expose.

Company-specific methodology, engineering memory, integration, and search capabilities can create substantial value for a design organization even when it never sells them as software.

Security can favor keeping the capability and its data inside an organization-controlled environment. FedEDA is designed to avoid centralizing raw circuit data because that data can be too sensitive to share. [11] Japanese local-LLM programs likewise keep design knowledge inside controlled environments, and incumbents and startups increasingly support on-prem or organization-controlled deployment.

The architecture doesn't dictate how ownership of the control plane is divided. Public evidence shows incumbent-provider, design-organization, independent-provider, and hybrid arrangements. Which arrangement captures the most long-term value remains open.

3.2 Where Technical Capital Can Accumulate

As capable foundation models become more widely available, durable advantage in AI-driven EDA may depend less on the model itself and more on what an organization builds around it. If intelligence gets cheaper, what stays scarce? The harder-to-copy assets may be the knowledge and systems an organization has built over time and continues to improve through use.

Strategy and capability-building research has long treated technical capability and organization-specific knowledge as accumulated organizational assets. [63, 79, 80] Organizational memory, reusable design rationale, and tacit engineering knowledge explain how these assets become more valuable through repeated use within one organization. [62, 81, 82] Current EDA accounts from Synopsys, Diamantidis, and Andersen describe company-specific methodology, accumulated domain experience, preserved rationale, and learning loops. [70, 71, 73, 91]

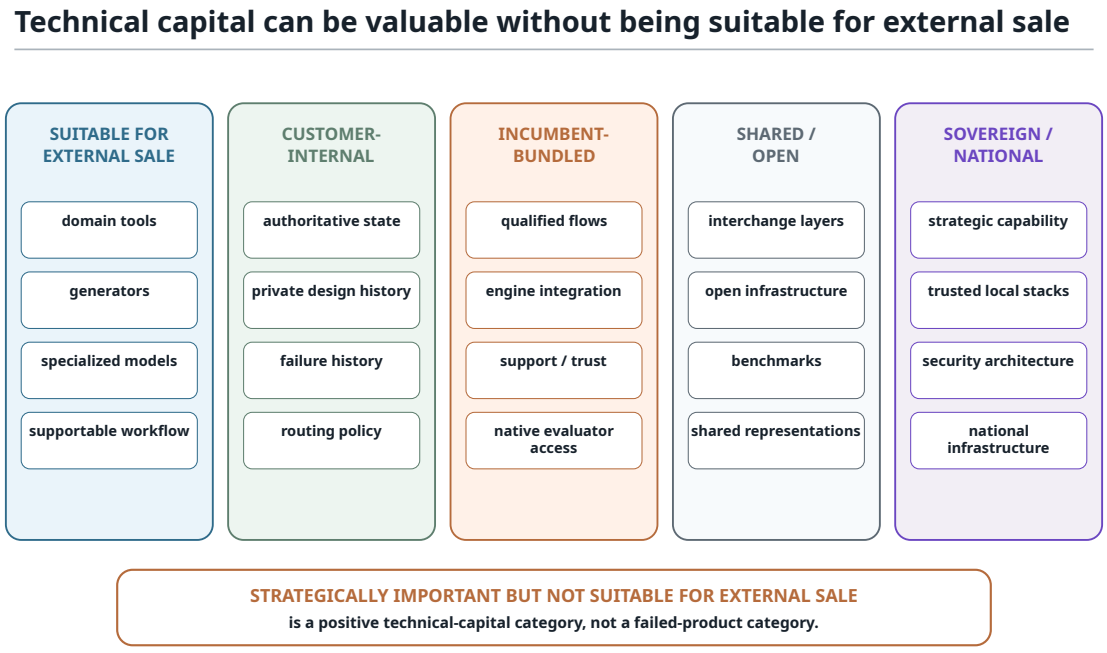


Figure 8. As capable foundation models become more widely available, durable advantage will depend even more on technical capital accumulated in design organizations, incumbent providers, foundries, ecosystems, and national programs. Much of that technical capital creates value primarily by improving the holder's own engineering capability rather than by being sold separately.

Technical-capital asset	What sustains its value	Transferability / market implication
Foundry qualification and physical models	These models take substantial process knowledge and qualification work to build, and their value depends heavily on the specific manufacturing process and on the confidence engineers can place in them.	Because these models depend on a specific manufacturing process and on qualification accepted by the engineers who use them, their value is difficult to transfer away from the foundry or established provider that maintains them.
Proprietary design history	It records failed experiments, important exceptions, constraints that proved consequential, and the reasons the design flow changed over time.	Its value is often greatest inside the design organization that created it because the history is proprietary and tightly tied to that organization's architecture and methodology. That makes it difficult to package as a transferable external product.
Workflow state and system of record	It keeps track of which design state is current, which constraints have been accepted, which methodology and tool settings produced the results, what the evaluators reported, where those results came from, what experiments failed, and who has authority to make the next decision.	The organization that maintains the authoritative design state and its history can retain an advantage even as the agent layer becomes easier to reproduce.
Tool integration	Its value comes from reliably connecting complex environments, proprietary data formats, security boundaries, and design methodology that's accumulated over years.	When an integration reaches deeply into proprietary formats, security boundaries, and long-lived methodology, and is difficult for others to reproduce, the organization that built and maintains it can retain a durable advantage.
Evaluator access	Access to trusted evaluators lets a system check proposed actions against reliable engineering evidence and use stronger checks when the stakes require them.	Reliable access to trusted evaluators can remain a source of advantage even as capable foundation models become more widely available.
Domain-specific representations	Representations such as XLS, HLS, generators, and other domain-specific abstractions encode design structure explicitly, so the system can work with meaningful design choices instead of searching over unconstrained low-level changes.	These representations can create substantial engineering value. It remains uncertain whether they can support a durable external product or business.

Technical-capital asset	What sustains its value	Transferability / market implication
Reusable generators and skills	Parameterized design assets and validated skills can be reused and improved across projects. Trace2Skill shows verifier-guided later-task improvement within its task regime; Dr. RTL establishes fold-isolated cross-design reuse; ChipMEM adds transfer from a frozen library to tasks not used to build that library. [10, 20, 98]	Repeated use can make these assets more valuable, but each result has an applicability boundary. Cross-tool, cross-node, and cross-organization transfer, requalification burden, ownership, maintenance, and negative transfer remain open.
Failure histories	Failure histories can keep the system from repeating unproductive experiments, help it choose appropriate validation, and signal when stronger evidence or human review is needed.	Failure histories often depend on proprietary architecture, internal methodology, and local context. Moving them outside the organization can expose proprietary information, and without that context failure histories are often much less useful.
Security and local deployment	For an externally supplied AI-driven EDA system, security requirements can determine where the system is allowed to run. Proprietary design information may be acceptable in infrastructure controlled by the design organization but not in a provider-controlled cloud service.	For some design organizations, support for on-prem or organization-controlled deployment can be a requirement for buying and using the product. A provider that can't meet those security requirements may be excluded regardless of the system's technical capability.
Support and trust	When engineering organizations choose a provider, the value comes not only from the algorithms but also from accountability, accumulated knowledge of corner cases, and a proven history of support.	Trust built through repeated successful use, responsive support, and accountability can take years to establish and is difficult for a new provider to reproduce.

The technical-capital assets in the preceding table matter because capable foundation models may become more widely available while other sources of advantage remain difficult to reproduce. Durable value can remain in authoritative engineering evidence, authoritative design state, proprietary history, integration, security, support, trust, and reusable knowledge that stays valid as tools and processes change. The important distinction is between creating value and being transferable across organizations. Across these asset classes, five economic behaviors recur.

Asset behavior	Economic consequence
Becomes more valuable through repeated use within one organization but transfers poorly	High internal value doesn't guarantee that an asset can be sold to other organizations.

Asset behavior	Economic consequence
Transfers readily across organizations but can lose differentiation as similar offerings become widely available	Broad reuse doesn't necessarily create durable pricing power.
Requires qualification	Moving the asset to a new process, project, or operating environment often requires requalification before engineers can rely on it.
Makes it costly for customers to switch providers	If customers incur meaningful costs when switching providers, the asset can help support a durable external business.
Serves as shared infrastructure	Shared infrastructure can create substantial value across an ecosystem. That value doesn't necessarily support a standalone product or a durable advantage for any one vendor.

Simply accumulating more data doesn't by itself create durable advantage in this sector. For any candidate asset, five ownership and reuse questions matter:

Decision dimension	Diagnostic question
Favored ownership structure	Which ownership structure do current conditions favor?
Value from reuse	Does the asset become more valuable as it's reused on more designs, incorporates more failure history, or is used by more organizations?
Transferability	Can the asset be transferred to another organization without exposing proprietary information or losing much of its usefulness?
Requalification	Does the asset require requalification when the tools, processes, or operating conditions it depends on change?
Value capture	Does the asset create pricing power for a provider or improve productivity inside the organization using it?

Open and public research institutions can accumulate technical capital through shared infrastructure, evaluated artifacts, maintained interfaces, and repeated engineering use. The next two cases examine how that accumulation works under different institutional arrangements.

CASE STUDY 6 — OPEN-RESEARCH ECOSYSTEM TECHNICAL CAPITAL: iEDA, AiEDA, and ECOS

Dimension	Finding
Technical system and accumulation mechanism	iEDA provides an open physical-design infrastructure, public tools, common data structures, and interfaces for building additional EDA capability. AiEDA turns design-flow results into structured data and vector representations for repeatable AI research and evaluation. ECOS extends the active lineage through open IP, SoC, PDK, education, and RTL-to-chip integration. The demonstrated accumulation path runs from design-flow results to machine-usable evidence and then into AI-aided-design work. Repeated design, teaching, debugging, and tapeout activity creates further opportunities to improve the system. [16, 42, 109, 154, 156, 202]
Engineering evidence and maturity	Engineering reality is established at several levels. The current iEDA homepage reports five tapeouts—two at 110 nm and three at 28 nm—and eleven tools, while the current repository and a 2024 peer-reviewed project-team paper report four; a 2023 project-team paper reports that the first 110 nm chip had booted. A 2026 conference abstract by an ECOS engineer reports fabrication of a 55 nm 100 MHz RetroSoC and use of ECOS Studio in One Student One Chip. The evidence closes returned-silicon status only for the first iEDA chip and leaves tool coverage for later tapeouts and production qualification of the current open PDK unresolved. [42, 154, 155, 202, 204]
Ownership and control	Public code, repositories, documentation, issue and contribution paths, examples, and part of the design data are reusable outside one firm. Operational control remains distributed and unequal: maintainers control releases, the complete paper-scale dataset isn't yet public, foundries control process collateral, and commercial evaluators remain authoritative at parts of the flow. Current ECOS documentation says that the ECOS Team combines the original iEDA and One Student One Chip teams. This structure supports the paper's label "technical capital accumulated in an open research ecosystem." Legal ownership and formal ecosystem-wide governance remain asset-specific or undocumented. [16, 42, 109, 155, 156, 202, 203, 204]

Dimension	Finding
Evidence	The evidence combines live project sites and repositories, peer-reviewed project-team papers, a 101 GB public iDATA release, current ECOS and ICsprout55 documentation, and a 2026 conference abstract by an ECOS engineer. Together, these sources establish active technical infrastructure, research use, silicon-facing work, and a continuing organizational lineage. Independent production adoption and commercial scale remain open questions. [16, 42, 109, 154, 155, 156, 202, 203, 204]
Demonstrated engineering value	iEDA and AiEDA provide maintained infrastructure, common interfaces, design-to-vector machinery, examples, evaluators, and a public research dataset; the surrounding ecosystem adds education, reusable design materials, and repeated silicon-facing work. These assets lower the setup cost for experiments, make results easier to reproduce and compare, and allow researchers to build on a common technical base. The evidence demonstrates engineering and research value. Financial returns and broad industrial adoption haven't been measured. [16, 42, 109, 154, 155, 156, 202, 204, 205, 206]
Potential durable advantage	Open code and data spread capability, while persistent interfaces, accumulated examples, tested flows, and maintainer know-how can make the ecosystem faster at running and interpreting the next experiment. Durability depends on maintenance, requalification as tools and PDKs change, portability across design contexts, institutional continuity, and renewal of tacit knowledge. Open diffusion broadens ecosystem value while making exclusive capture by any single participant less likely. [16, 42, 109, 154, 156, 201, 202, 203, 204, 205, 206]
What remains open	Formal decision rights across OSCC and ECOS assets; identities, returned-silicon status, and tool coverage for later tapeouts; completion and terms of the full approximately 600 GB iDATA release; maturity of AiEDA beyond its current iEDA-centered implementation; the boundary between open flows and commercial or foundry-controlled authority; independent external reuse; the strength of education-to-contribution and tapeout-to-tool-improvement loops; and where economic value is ultimately captured.

CASE STUDY 7 — PUBLIC-RESEARCH TECHNICAL CAPABILITY: QiMeng

Dimension	Finding
Technical mechanism	QiMeng is an evolving public-research umbrella for automated processor hardware and supporting software rather than a finished integrated system. QiMeng-CPU-v1 learns BSD-based RISC-V core logic from finite external input-output observations drawn from processor verification/reference behavior; the paper reports extremely high validation accuracy, but not exhaustive formal correctness over every possible input and state. QiMeng-CPU-v2 tackles a different problem: State-BSD predicts inter-instruction dependencies from internal processor state, bypasses dependencies it can't predict safely, and uses SMT to verify predictor precision. Work now presented under the broader QiMeng umbrella also includes automated OS configuration and neural-symbolic transcompilation, some of which predates the umbrella framing. [41, 157, 159, 161, 162] The public portfolio also includes repositories for QiMeng-CPU-v1, QiMeng-Kernel, and QiMeng-MuPa. [198, 199, 200]
Evaluator regimes and evidentiary limits	QiMeng-associated projects use different evaluator regimes. QiMeng-CPU-v1 is learned from finite input-output examples and evaluated with large validation suites. The paper uses the Pentium 4 verification regime to motivate a target error on the order of 10^{-13}; it doesn't report that threshold as a measured end-to-end validation result or an exhaustive formal proof of whole-CPU behavior. QiMeng-CPU-v2 formally checks State-BSD predictor precision with SMT and also uses commercial simulation, FPGA execution, Linux, SPEC, Dhrystone, and CoreMark for behavioral and performance evidence. The researchers explicitly retain conventional EDA in the broader automation architecture for implementation, evaluation, and verification. [157, 160, 161]
Institutional locus and control	Under the current institutional arrangement, the hardware research is centered in public CAS research organizations, with software-side collaboration and repeated grant support documented in published work. This supports "public-research technical capability" as an institutional-locus description, not as a complete legal-ownership map. Publications and repositories make some methods, code, and designs externally available. Repeated research likely creates tacit integration and debugging knowledge in the participating institutions, but that tacit asset is inferred rather than directly inventoried. [41, 160, 161, 196, 197]

Dimension	Finding
Evidence	The QiMeng-CPU-v1 project-team paper reports generating the BSD-based CPU core in about five hours on a 68-server cluster; that timing excludes the rest of the SoC and backend circuit design. The generated logic was subsequently implemented and taped out, and the authors report Linux operation and Dhrystone performance comparable to Intel 80486SX. QiMeng-CPU-v2 is a roughly 17-million-gate four-ALU superscalar RISC-V-32IA design. Its project-team paper reports FPGA validation with Linux and SPEC, Dhrystone/CoreMark evaluation, and a Dhrystone-based comparison near Cortex-A53; at publication, 28 nm tapeout was in progress. The 2025 umbrella paper reported that QiMeng was still in phase one, with comprehensive integration future work; a fresh 2026 search found newer component papers but no primary source closing that integration gap. AutoOS and QiMeng-Xpiler show software-side breadth in work now grouped under the umbrella. [157, 158, 159, 161, 162]
Demonstrated engineering value and economic limits	Public evidence demonstrates substantial task automation and measured technical results, not a complete accounting of total engineering labor or economic value. The v1 authors report about five hours for BSD-based CPU-core generation but explicitly exclude SoC/backend work and use substantial cluster compute. AutoOS reports up to roughly 25% performance improvement over vendor-provided configurations in tested settings. QiMeng-Xpiler reports about 95% average translation accuracy across evaluated cases, while its computation accuracy varies by direction and expert-optimized vendor libraries remain stronger on average performance. These results show useful automation in bounded workflows; they don’t establish program-level dollar ROI, national economic return, total labor savings, external pricing power, or commercial reproduction cost. [157, 159, 161, 162]
Potential durable advantage	The observable portfolio extends beyond any one generated CPU: design methods, evaluation and search machinery, code, processor artifacts, benchmarks, and public research artifacts accumulate across the line of work. Public release makes some of those assets portable and available for external reuse, but meaningful diffusion outside the originating research network isn’t established by availability alone. Repeated debugging, integration, and fabrication work likely creates tacit know-how, while unreleased data/training assets and specialized evaluation access remain less transferable; those tacit and transferability claims should be treated as analytical inferences unless directly evidenced. [41, 159, 162, 198, 199, 200]

Dimension	Finding
Alternative institutional structures	The same kinds of methods and artifacts could be organized differently. An EDA incumbent could integrate processor-design automation with its trusted tools; a chip company could internalize the methods and keep design context private; an independent company could commercialize the automation; an open ecosystem could distribute reusable artifacts. QiMeng makes another institutional path concrete: public research can develop and retain a broad portfolio without requiring a standalone software vendor. Whether that portfolio compounds into durable advantage across future processor families remains open.
What remains open	Public evidence doesn't yet establish completed 28 nm fabrication or post-silicon measurements for QiMeng-CPU-v2, a finished integrated/self-evolving QiMeng system, complete public release of the data and training infrastructure behind every component, meaningful independent external reuse at scale, legal ownership/governance of all umbrella assets, or program-level total-cost and economic returns. It also remains unclear how much of the accumulated portfolio transfers to future processor families without substantial revalidation, retraining, or redesign.

Open-research and public-research arrangements shape where technical capital accumulates and who can maintain, qualify, and reuse it. The next question is how evaluated engineering experience becomes a reusable asset across later tasks and designs.

3.3 Engineering Memory: From History to Reusable Skill

Recent evidence shows that engineering experience can be preserved, validated, and reused across later tasks or designs. That establishes engineering memory as a technical asset. Whether a provider can turn that reusable knowledge into a durable business advantage remains unresolved.

A stateless assistant sees a prompt, produces an answer, and stops. A persistent engineering system can preserve what happens from proposal through evaluation and decision, record why a candidate succeeded or failed, turn that experience into a validated reusable skill, and apply the skill to a later design.

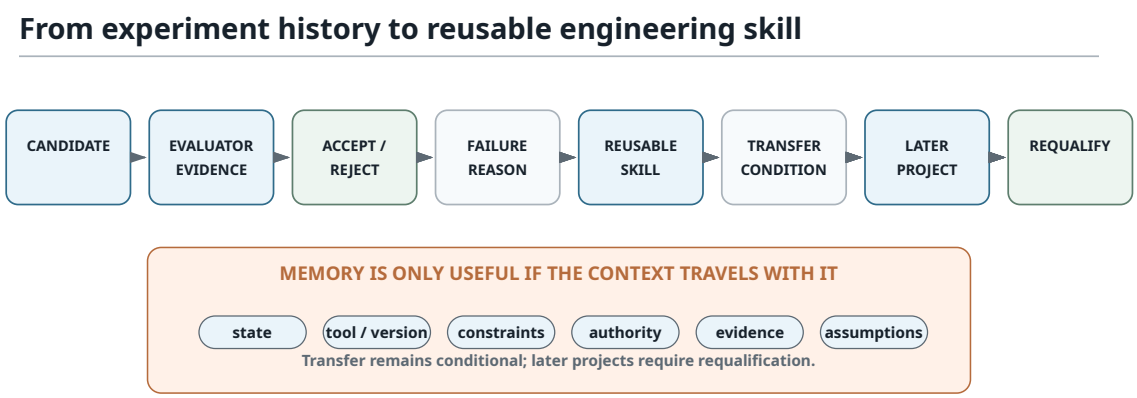


Figure 9. Evaluated engineering experience can become a reusable technical asset only when it's preserved together with the relevant design state, tool versions, constraints, decision authority, evidence, and assumptions. Whether that knowledge can transfer across organizations or produce durable business advantage remains unresolved.

Trace2Skill demonstrates that execution traces and verifier feedback can be used to produce explicit skills that improve later agent behavior. [10]

ChipMEM distinguishes a library that continues learning from a frozen library applied to tasks not used to build it. Its verification-grounded skills improve accepted outcomes in small experiments, while broader transfer, runtime, and long-term validity remain open. [98]

A tool-interactive RTL-to-GDS benchmark reaches a complementary result: structured interfaces, persistent context, and skills can improve local completion, yet similar milestone scores can carry very different runtime and token costs. [99]

Recent research on engineering memory distinguishes several evidence states. The following table separates them by mechanism, transfer test, and governance so endpoint success doesn't stand in for evidence quality.

Evidence pattern	What the source tests	What the evidence supports
Frozen skill transfer	ChipMEM compares an evolving skill library with transfer from a frozen library to EDA tasks withheld from skill development. [98]	Shows bounded transfer to tasks not used for skill development; broader task, organization, and time horizons remain open.
Frozen meta-skills	Meta-skills construct tools, state, and controllers for tasks withheld from meta-skill development. [119]	Separates reusable support-design knowledge from a finished task solution.
Capability-grounded transfer	EngramBench separates capability overlap from solution reuse with frozen deliveries and hidden evaluators. [120]	Supports a transfer protocol within a small non-EDA experiment.

Evidence pattern	What the source tests	What the evidence supports
Memory with continued adaptation	SAGE uses hardware-tested NPU-kernel outcomes while continuing to update memory on a new library. [121]	Demonstrates accumulation and adaptation, rather than frozen unseen-task transfer.
Construction-order sensitivity	RoutePrism changes the order of identical source records and tests which evidence survives. [122]	A larger memory can still lose decision-relevant evidence.
Learned policy parameters	PCPO moves accumulated search experience into model parameters. [115]	This is a different asset from an explicit, inspectable skill library.
Adaptive skills and harnesses	Vestrum, EASE, SkillSpec, GenMem, CodeSkill, SkillVine, Beyond the Model, and Component-Aware Feedback test different ways to update, gate, represent, branch, and evaluate skills. [124, 125, 126, 127, 128, 129, 130, 131]	Maps distinct mechanisms in software and memory benchmarks; transfer to EDA requires separate validation.
File- and source-preserving memory	CAMG, MemCodex, and MemFit study file-based, layered, or append-only memory. [137, 138, 139]	Lower write cost or source retention can move work to retrieval, summarization, and recombination.
Authorization-bound memory	Audience-bound memory carries read authorization with stored and derived memories. [132]	Read authorization doesn't settle write integrity, action authority, ownership, or reuse rights.

Dr. RTL provides semiconductor-specific evidence that reusable skills can transfer across designs. The study evaluates a library of 47 interpretable pattern-strategy entries on 20 real-world RTL designs in an industrial EDA workflow. Relative to an industry-leading commercial synthesis tool, it reports average WNS improvement of 21%, average TNS improvement of 17%, and average area reduction of 6%. [20]

Engineering experience can produce reusable EDA skills that transfer across designs in controlled research settings. DSO.ai adds a bounded commercial example: Synopsys reports that reuse of a 5 nm training database at 4 nm reduced a related-node Fusion Compiler retargeting workload sixfold versus cold start. [20, 91] These results establish reuse across specific designs and related nodes, while broader transfer and durable provider advantage remain separate questions. Five conditions affect whether engineering memory continues to generate value:

Memory condition	Why it matters
Transfer scope	A skill that transfers across blocks within one benchmark or design organization may not transfer to different architectures, process nodes, tool versions, or organization-specific methodologies.
Validity over time	Libraries, PDKs, constraints, and tool behavior change. A skill that was valid under earlier conditions may need to be revalidated when those conditions change.
Ownership	A design organization may treat failure history and methodology as proprietary information. An external provider may manage the memory system without owning that information or having the right to reuse it for other customers.
Privacy	Cross-organization learning can create value, but it can also expose proprietary information. FedEDA shows that privacy constraints in EDA can motivate federated learning rather than centralized data sharing. [11]
Validation	Stored engineering experience becomes reusable only when it records the conditions under which a lesson is valid, the evidence and provenance supporting it, and what changes would require the lesson to be revalidated.

Engineering memory doesn't have to be encoded in the base model. Skills, playbooks, constraints, and evidence histories can remain explicit and inspectable rather than being represented only in model weights.

Keeping engineering memory explicit and inspectable raises five governance questions:

Governance dimension	Question
Learning	What did the system learn?
Evidence	What evidence supports the stored knowledge?
Applicability	Under what conditions does the stored knowledge apply?
Approval	Who approved the stored knowledge for reuse?
Expiration	When should the stored knowledge no longer be treated as valid for reuse?

Engineering memory becomes an explicit engineering artifact with defined evidence, applicability, approval, and validity.

CASE STUDY 8 — ENGINEERING MEMORY: Three Tests of Transfer

Engineering memory matters when an evaluated result improves a later engineering decision. Three systems test increasingly demanding forms of reuse relevant to transfer: Trace2Skill evolves a task-specific natural-language skill from verifier-guided traces; Dr. RTL tests a pattern–strategy library across designs; and ChipMEM tests a frozen library on unseen tasks. [10, 20, 98]

Evidence regime	What was tested	What it establishes
Trace2Skill	Repeated rollouts on difficult Verilog tasks were distilled into task-specific natural-language skills, with sanitized verifier feedback available during skill evolution.	Verifier-guided later-task improvement without weight updates. Frozen-library, cross-design, cross-tool, and cross-organization transfer remain untested. [10]
Dr. RTL	Dr. RTL's 47-entry library captures evaluated synthesis experience. In the four-fold skill-generalization test, the library was reset for each fold, skills were extracted from 15 designs, and applied to five unseen designs under an industrial EDA workflow.	The fold-isolated evaluation establishes cross-design skill reuse on unseen designs. The full Dr. RTL system reports average improvements of 21% in WNS, 17% in TNS, and 6% in area compared with an industry-leading commercial synthesis tool. [20]
ChipMEM	Procedural memories admitted after synthesis, simulation, or formal checks were tested with evolving and frozen libraries on tasks that weren't part of the initial library-development set.	Transfer from a frozen procedural memory library to new testbench-generation tasks. ChipMEM produced accepted testbenches for all 20 CVDP tasks that weren't used to build the library, versus 18 of 20 without a procedural memory library; it also reports 39 of 54 versus 35 of 54 equivalence-passing outputs overall. Each configuration was evaluated once. [98]
Industrial corroboration — Synopsys DSO.ai	A 4 nm retargeting run was warm-started from prior search knowledge developed on a related 5 nm RISC-V design.	Provider-reported related-node reuse: 15 jobs in one iteration rather than 30 jobs over three iterations for the cold start, described as a sixfold reduction and one day with zero human intervention. The demonstrated boundary is related-node retargeting. [91]

The evidence forms a ladder: improvement within a task regime, reuse across designs, and frozen reuse on unseen tasks. The DSO.ai example adds bounded production evidence for a related-node warm start. Each step expands the applicability claim and the requalification burden.

Decision question	Case finding
What's the memory artifact?	The reusable asset may be a natural-language skill, a pattern–strategy entry, a verified procedure, outcome statistics, or qualified search state. Its value depends on the artifact, its provenance, its applicability boundary, and the evidence that admitted it.

Decision question	Case finding
What does the evaluator establish?	Synthesis, simulation, formal checks, or a task verifier establish that a result met specified criteria under a particular configuration. Future reuse requires separate validation of the extraction method, applicability boundary, and authorization.
Where can the memory transfer?	Applicability must name the transfer boundary: later task, design family, constraints, tool and version, PDK and libraries, process node, evaluator, organization, and security domain.
When must the memory be requalified?	Changed tools, libraries, constraints, architectures, design populations, evaluators, or authorization boundaries can invalidate prior lessons. New outcomes determine whether memory is retained, revised, requalified, or retired.
Who controls the memory?	Control can be split among the design organization, model or service provider, EDA platform, and shared infrastructure. Practical control belongs to the parties that can inspect, validate, update, revoke, export, and reuse the artifact and its underlying evidence. Bargaining power follows evaluator access, qualified-flow integration, and reuse rights; switching costs rise when organization-specific memory can't move.
How can qualified memory create economic value?	Qualified memory improves routing by raising the expected value of the next transformation, evaluator, or search direction. Reuse can reduce repeated search and evaluator calls, shorten retargeting, preserve scarce expertise, and improve later decisions. Public evidence measures technical and workflow outcomes. Revenue and customer-retention effects remain unmeasured.
What evidence is still needed?	Independent replication, larger-scale durability, negative-transfer rates, maintenance burden, and cross-tool and cross-node portability remain open. [119, 120, 122] Cross-organization reuse requires explicit authorization, provenance, and security controls. [132] The distribution of value among customers, EDA platforms, and independent providers also remains open.

A reliable engineering memory system treats reuse as a controlled lifecycle.

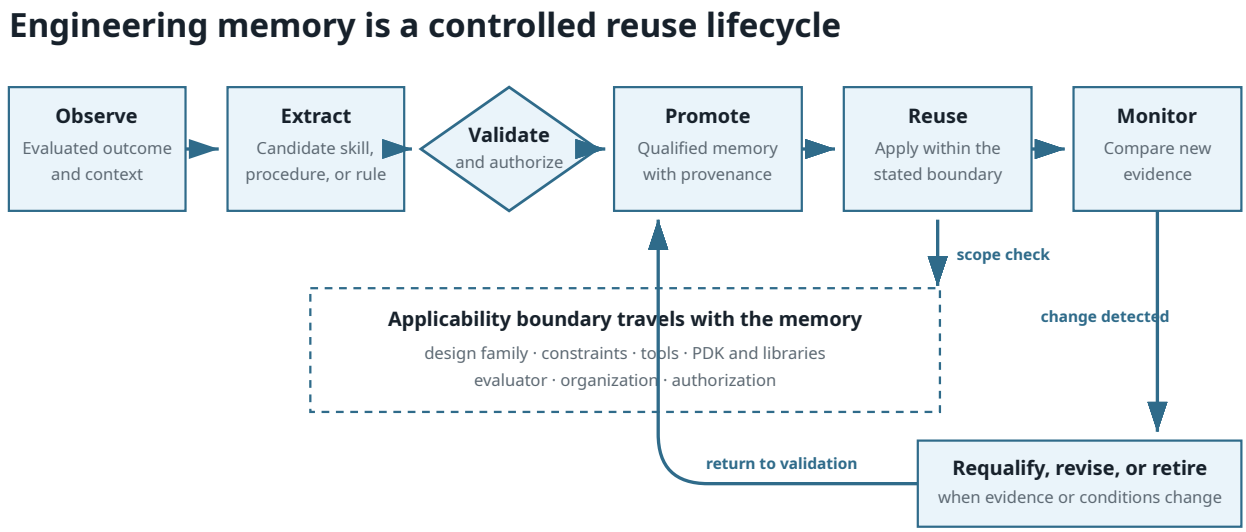


Figure 10. Engineering memory becomes durable when an evaluated lesson meets explicit criteria for reuse, travels with its supporting evidence and applicability boundary, and is requalified when conditions change.

Engineering memory becomes an economic asset when the organization can tell which lesson applies, who may use it, which evaluator supplied the supporting evidence, and what changes require requalification. Storage creates potential; governed reuse creates cumulative advantage.

Persistent engineering memory depends on a reliable system of record. An agent can't reliably learn from prior work unless the organization can identify which design version, constraints, tool settings, and evaluator results were authoritative at the time.

Low-cost computational search creates strategic value only when an engineering system can obtain authoritative evidence and preserve what it learns across iterations. It must also build reusable assets that remain valid and can be used or transferred without violating qualification, security, or IP constraints.

4. Similar Technical Loops, Different Institutional Structures

4.1 Technical Capital Across Ecosystems

Chip design across ecosystems relies on a recurring engineering loop: systems propose or search, authoritative evaluators produce evidence, organizations preserve what was learned, and the result changes the next action. The loop operates inside commercial platforms, internal design organizations, open infrastructure, public research programs, universities, and national capability initiatives.

Institutional settings differ in who controls evaluators, owns data and interfaces, maintains tools, carries qualification costs, and retains engineering memory. Those control and ownership choices determine which assets accumulate, who can reuse them, what travels across tools or organizations, where dependence forms, and who can capture value.

Technical capital can accumulate in incumbent platforms, internal design organizations, independent providers, open infrastructure, public research programs, and reusable engineering-memory systems. Across ecosystems, the underlying engineering loop recurs even as ownership and institutional organization differ.

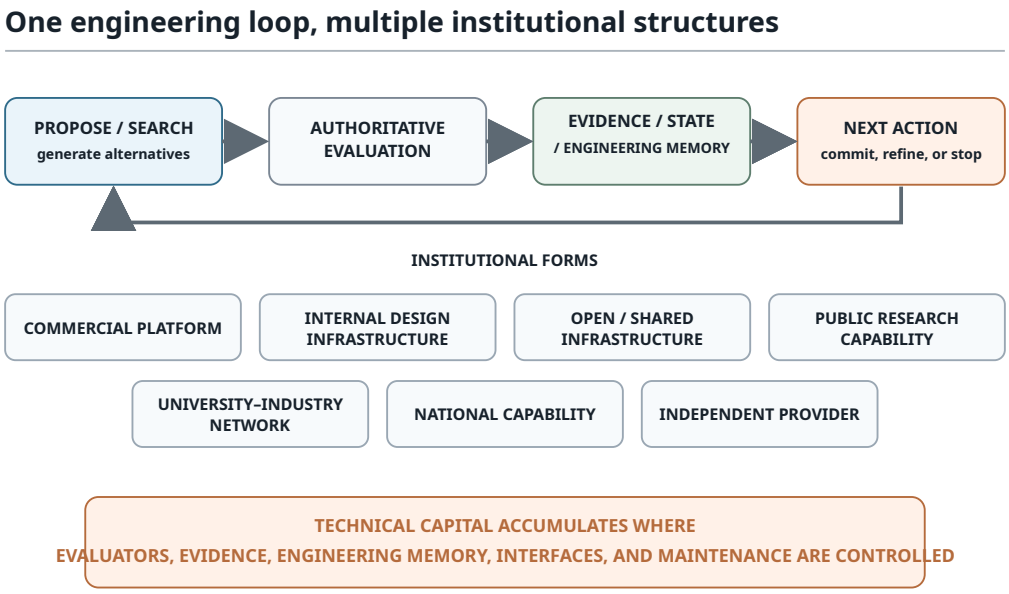


Figure 11. The engineering loop recurs across institutional settings. Ownership of evaluators, evidence, engineering memory, interfaces, and maintenance determines where technical capital accumulates and how it can be transferred or commercialized.

Representative evidence from ten ecosystems shows how the recurring loop is embedded in different institutional structures:

Ecosystem	Representative evidence	Institutional form of technical capital
China — PRC	iEDA/AiEDA/ECOS open design infrastructure and education/tapeout networks; QiMeng processor-design and supporting-software research. [16, 41, 42, 109, 157, 159, 160, 161, 162, 196, 197, 202, 204]	Open research infrastructure and public-research capability distribute control across maintained tools, shared interfaces, evaluators, repositories, research institutions, and repeated engineering use.
China — Taiwan	National Taiwan University and MediaTek-linked research on AI for EDA, physical design, and AI-assisted chip design. [43, 207]	University–industry and foundry-adjacent networks accumulate technical capital through shared research, integration expertise, and commercialization channels.
France	CEA A-DECA and VPSim; SoC Planner; LIP6 CAIRO/OCEANE/Coriolis; Defacto and Innova. [13, 30, 32, 109, 114, 182]	Public research infrastructure, independent vendors, workflow-data systems, and incumbent ownership connect long-lived automation machinery to commercial design flows.
Germany	Fraunhofer Chip AI and FRAUKE programs for AI-assisted chip design, analog automation, verification, and certification. [44, 208]	Public research and industrial collaboration organize technical capital around assured deployment, domain expertise, and reusable engineering

Ecosystem	Representative evidence	Institutional form of technical capital
		infrastructure.
India	ChipIN/DLI shared EDA infrastructure and CaretEDA's commercial open-stack program. [45, 209]	Publicly supported tool access and independent open-stack commercialization combine ecosystem capacity with a product and services path.
Israel	National AI and semiconductor-design strategy and vendor-neutral post-silicon validation infrastructure. [47, 211]	National capability programs coexist with independent service platforms built around integration, validation, and semiconductor engineering talent.
Japan	GenAI and HLS programs, local-model-aided design, analog knowledge formalization, and established HLS and design-automation traditions. [14, 110, 167, 168]	National programs, local deployment, tacit engineering knowledge, and long-lived automation machinery keep technical capital close to controlled design environments.
South Korea	Rule2DRC; university work in physical prediction, differentiable optimization, analog sizing, and synthetic data; privacy-preserving federated EDA. [7, 11, 181]	University research, execution-grounded evaluation, and privacy-preserving collaboration make evaluator access and data governance central to accumulated capability.
United Kingdom	National AI-hardware planning and proof-centric RTL-agent research represented by APRIL/PRAMANA. [46, 210]	Public capability programs and research networks organize technical capital around formal assurance, evidence, and reusable design capability.
United States	Incumbent EDA platforms, corporate design-automation research, open compiler and EDA infrastructure, and AI-native providers. [1, 2, 3, 4, 6, 15, 36, 38, 39]	Commercial platforms, internal design infrastructure, open projects, and independent providers compete over orchestration, evaluator integration, engineering memory, and workflow control.

Technical capital can accumulate in a commercial platform, an internal design environment, an open stack, a public research portfolio, a university–industry network, or a national capability program. Each institutional form assigns control of evaluators, evidence, engineering memory, interfaces, and maintenance to different actors, shaping access, switching costs, transfer, qualification, and the path from engineering capability to durable economic value.

5. Value Capture and Market Structure

5.1 Build, Buy, or Keep Internal?

The ownership structures described here will change as technical and economic conditions change. Security, internal-build capacity, integration burden, evaluator access, methodology importance, transferability, and platform economics all affect whether a capability is built internally, bought, integrated into a broader platform, shared, maintained as a national capability, or offered as an independent product. Watching those forces over time can show when a different ownership structure is becoming more attractive.

Choosing whether to build, buy, or keep a capability internal is an economic as well as technical decision. Internal ownership becomes more attractive when the capability becomes a valuable organization-specific resource, creates costly dependence on external suppliers, is cheaper to coordinate within the organization, or secures complementary assets needed to capture value. [65, 77, 78, 83]

Current Synopsys–Amazon and AWS–ChipAgents materials link cloud architecture, IP access, private execution environments, and orchestration. [134, 135] They establish current collaboration patterns while leaving AI-attributable economics, ownership of engineering memory, and switching cost open.

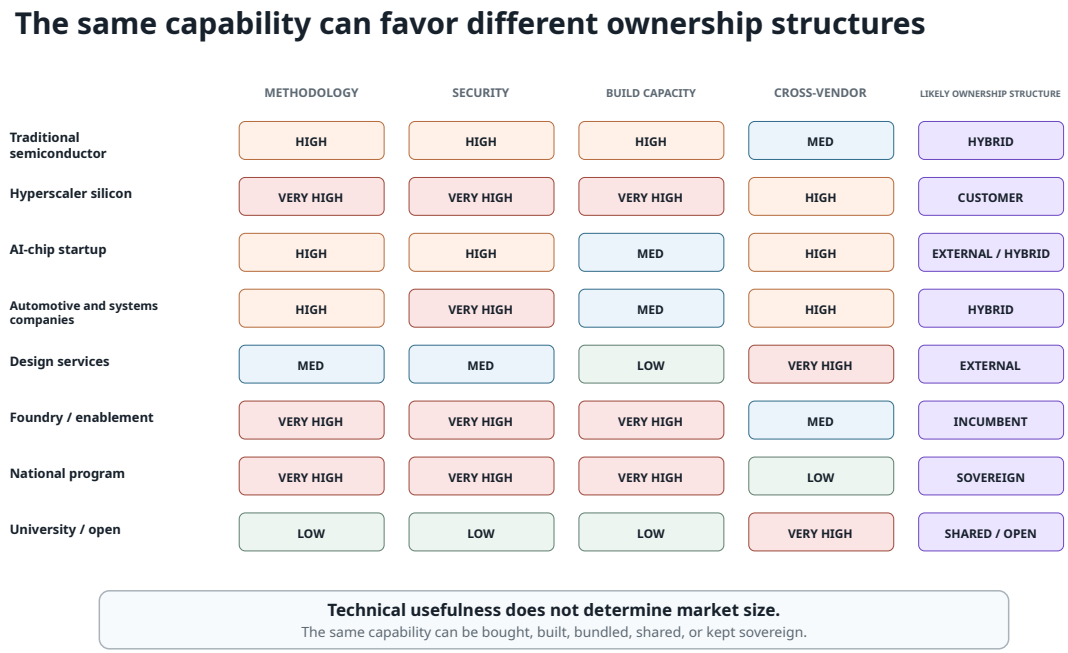


Figure 12. The same capability can be owned and supplied in different ways as conditions change. Security, methodology importance, internal-build capacity, cross-vendor need, and willingness to share learning influence whether it's bought, built internally, integrated into a broader platform, shared, or kept private. Those choices also affect the economics of offering it as a standalone product.

Buyer type	Distinctive objective / constraint	Likely ownership structure
Traditional semiconductor companies	Traditional semiconductor companies need compatibility with established flows, qualified results, mature methodology, and low integration risk. Internal CAD teams may retain substantial control over workflow and methodology.	Often hybrid: companies buy established EDA engines while retaining important methodology and workflow capabilities internally.
Hyperscaler and platform silicon teams	Large design scale and strong internal engineering capacity can justify substantial investment in internal design environments. Google and Apple provide direct public evidence. [5, 18, 183, 184]	These teams may buy commercial EDA engines while building the surrounding control, design-state, search, and AI-assistance layers internally.
Chip startups	Chip startups need to reach first silicon quickly, explore architectures efficiently, and make scarce specialist expertise go farther. They often have less capacity to build and maintain large internal EDA platforms.	Chip startups are likely to rely on external EDA tools or hybrid arrangements instead of building large internal EDA platforms.
Automotive and system companies entering silicon	Automotive and system companies entering silicon may have deep system expertise but less established EDA infrastructure. They need tools that are usable, predictable, and easy to integrate with software-development workflows.	These companies are likely to rely on integrated external EDA tools or hybrid arrangements instead of building extensive internal EDA infrastructure.
Design services	Design-service firms work across different customers and tool environments. Cross-vendor neutrality and repeatability are core requirements, while throughput and visibility into project state affect how efficiently they can deliver work.	Design-service firms are likely to favor vendor-neutral or hybrid orchestration that can work across multiple customer tool environments.
Foundries and enablement providers	Foundries and enablement providers need qualified implementation flows tied closely to process requirements and effective support for customers using those flows.	Foundries and enablement providers are likely to retain control over process-specific enablement and qualification while integrating commercial EDA and AI-assisted tools into qualified flows.
Government and national programs	Government and national programs may prioritize domestic technical capability, workforce development, and resilience over standalone software revenue.	These capabilities may be maintained directly by public institutions, supported through public-private arrangements, or combined with commercial tools and infrastructure.

Buyer type	Distinctive objective / constraint	Likely ownership structure
Universities and open ecosystems	Universities and open ecosystems generally prioritize accessibility, reproducibility, and the ability to experiment freely.	These environments often rely on shared or open infrastructure maintained by universities, public programs, consortia, or contributor communities rather than a single commercial owner.

Buyer type is only a rough guide. Ownership depends more directly on internal engineering capacity, security requirements, cross-vendor needs, methodology importance, and willingness to share engineering knowledge outside the organization. Those factors influence whether a capability is bought or built internally, used as a standalone capability or integrated into a broader platform, and shared or kept private. Usability also matters because EDA purchasing decisions depend on more than PPA.

A small PPA improvement may not justify weeks of engineering time, rare expertise, or a more fragile flow. The engineering decision depends on the business outcome under engineering and schedule constraints, with PPA as one input.

Objective dimension	What it measures / why it matters
Engineer-hours per accepted design improvement	How much engineering time is required to produce and validate a design improvement before it can be accepted.
Time to first good implementation	How quickly the flow produces an implementation worth carrying forward.
Onboarding	The effort required for a team to begin using the capability productively.
Diagnosability	Whether engineers can determine why a failure occurred and what to change next.
Reproducibility	Whether repeated runs under controlled conditions produce consistent results.
Failure recovery	How much effort is required to recover when the automated flow fails.
Verification burden	How much verification is required before generated or optimized output can be accepted.
Compute / license cost	The cost of compute resources and licenses required to obtain the engineering result.
Design optionality	How much useful design freedom remains as more design decisions are made.
Schedule risk	Whether the capability makes delivery timing easier or harder to predict.

Security can shift the build-versus-buy boundary. Sensitive design data, PDKs, layout data, and failure history can make cloud-hosted or provider-controlled engineering memory unacceptable even when the underlying capability is strong.

Conventional market analysis can miss an important category here: technical capital that's strategically valuable but not well suited to sale as a standalone product.

Technical importance and suitability for sale as a standalone product are different properties. A capability can create enormous value within an enterprise or for a nation while making little sense as a standalone software product.

Strategically valuable internal technical capital includes internal methodology, authoritative state, private history, evaluator policy, security architecture, and other capabilities often owned by the organization that controls the relevant context, holds the relevant authority, or bears the relevant risk. Changes in security requirements, transferability, evaluator access, or internal-build economics can shift the build-versus-buy boundary. A hyperscaler can create enormous value with an internal methodology agent without ever buying or selling it as a product.

5.2 Agentic EDA Economics and Industry Structure

Agentic EDA can change how tools are consumed even when the underlying engines stay the same. Human-driven flows typically use licensed seats operated by experts, with scheduled runs and manually selected experiments; machine-driven flows can consume the same tools very differently.

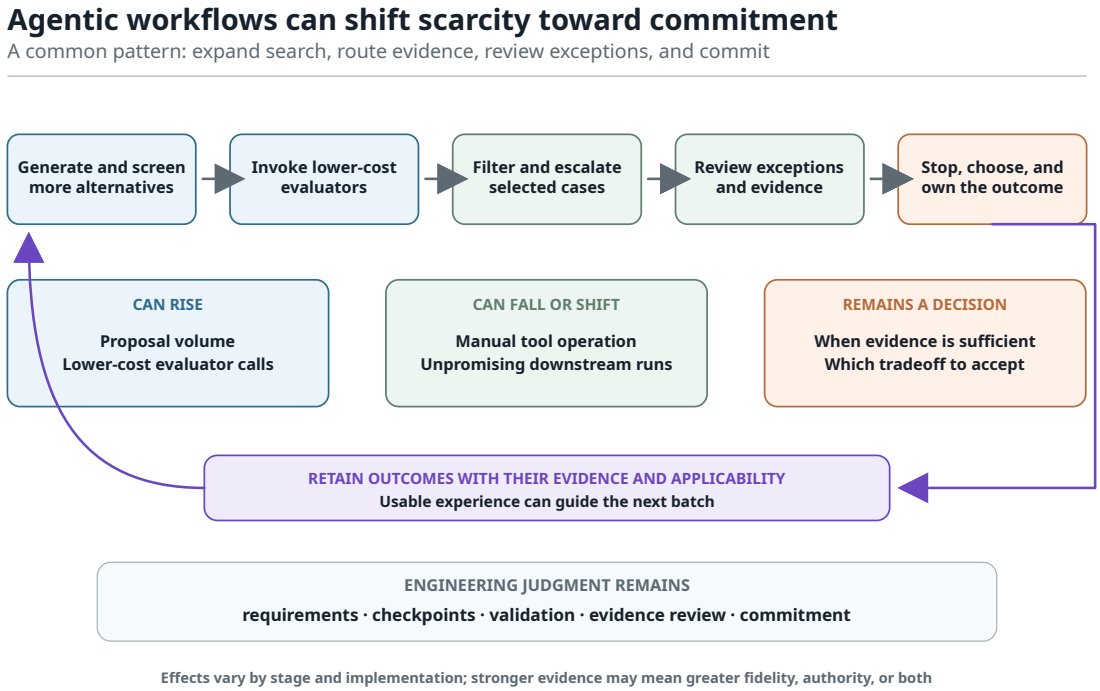


Figure 13. When agents automate much of the proposal-and-evaluation loop, less expert-engineer time is needed to run it, but evaluator use can increase. As proposing and evaluating alternatives become cheaper, the scarce resource can shift toward organizational commitment: when to stop, which tradeoff to accept, and who bears the consequences.

Loop step	Effect on evaluator use
Propose	Agents can generate many alternatives for evaluation.
Invoke	Each alternative can invoke simulation, formal verification, synthesis, HLS, timing analysis, physical estimation, or place-and-route tools.
Filter	Lower-cost evaluators can filter candidates.
Escalate	A subset of candidates can be passed to higher-cost evaluators when stronger evidence is needed.
Learn	Results from each batch can be retained and used to guide the next.

Machine-driven flows can increase evaluator use in some stages while reducing downstream use through filtering or substitution in others.

Spending pattern	Mechanism	Public evidence / observable effect
Expansion	Higher proposal volume can drive higher evaluator use.	NVIDIA and Cadence describe agent loops that run hundreds of simulations and formal checks, execute millions of verification tests, and consume very large amounts of compute. [4] Synopsys FlexEDA supports by-the-minute metering, large-scale parallel access, and budget controls. [12] Synopsys also reports contract uplift associated with DSO.ai customers. [179]
Filtering / substitution	A lower-cost inner loop can filter out candidates before they reach higher-cost downstream evaluators.	Open-source flows, HLS, virtual prototypes, or learned surrogates can filter candidates before higher-cost commercial signoff tools are used.

These patterns can coexist at different stages of a flow. Agentic operation can increase some forms of spending while reducing others or shifting where costs occur, so greater automation doesn't imply uniformly higher costs.

What can increase	What can decrease or shift	Why
Lower-cost compute	Expert engineering attention per alternative.	Automated loops can explore more alternatives without requiring expert engineering attention to increase proportionally.
Simulation and other lower-cost evaluator calls	Higher-cost signoff runs on candidates unlikely to meet signoff requirements.	Lower-cost evaluators can filter out candidates unlikely to meet signoff requirements before higher-cost signoff runs are performed.
Use of incumbent vendors' EDA tools	Engineer time spent on manual tool operation.	Automated loops can use incumbent vendors' EDA tools more often while reducing the engineer time spent operating them manually.
Usage-based spending on cloud compute, model tokens, or EDA tools	Share of spending on seat licenses and other largely fixed costs.	Automated operation can shift spending toward usage-based charges without necessarily increasing total spending.

Industry participants report that token efficiency and budget predictability are becoming operational considerations in agentic EDA. [21] The same technical architecture can be embedded in different industry structures.

Evidence state	What current public evidence supports
SUPPORTED	EDA is available under usage-based pricing models. Agentic workflows can generate high volumes of tool calls. Model and token costs are becoming explicit items in engineering budgets. At least one incumbent EDA vendor reports higher contract values associated with AI-driven optimization.
SCENARIO-DEPENDENT	Total EDA spending increases. Usage-based pricing broadly replaces seat-based licensing. A vendor-neutral orchestration layer becomes a major control point. Who controls usage metering becomes a central question in EDA industry economics.

Scenario	Ownership and control structure	Economic implication
Incumbent EDA vendors integrate the agent layer	The agent layer is integrated into incumbent platforms built around established engines and access to evaluators.	If comparable functionality becomes part of incumbent platforms, independent interaction layers may be less viable as standalone products. More of the economic value may then accrue to the incumbent platform and the existing workflow around it.
Vendor-neutral orchestration gains meaningful control over the workflow	An independent layer coordinates work across multiple vendors, evaluators, and systems used by design organizations.	A standalone market can persist if buyers continue to pay for vendor neutrality, integration, and orchestration instead of building those capabilities internally or obtaining comparable functionality as part of a larger platform.
Design organizations build internal silicon platforms and purchase EDA engines	The design organization develops its own control plane, methodology, search policy, and internal AI, maintains control of its design state, and licenses authoritative EDA engines from vendors.	Design organizations can create substantial value internally and continue to rely on commercial evaluators without needing a separate control-plane provider.
Design organizations use higher-level representation platforms as the primary authoring environment for some designs	An external provider controls the higher-level representation and compiler layer, while downstream implementation and signoff remain in separate EDA engines.	If design organizations adopt the higher-level representation and can manage downstream QoR and verification obligations, the company supplying that representation and compiler layer can capture more economic value.
Specialized vendors control important domain-specific workflows	Specialized vendors control workflows in verification, analog, DFT, post-silicon, and other domains, but authoritative evaluation remains with incumbent EDA tools or evaluators developed inside design organizations.	Specialized vendors can sustain standalone markets when it's uneconomic for broader platform vendors to absorb their domain expertise, integration, and workflow knowledge and for design organizations to reproduce those capabilities internally.
Open tools or approximate evaluators handle most design exploration	Exploration relies on open or low-cost search tools and approximate evaluators, but commercial EDA tools remain authoritative for downstream implementation and signoff.	As exploration becomes cheaper and more widely available, economic value can remain concentrated with vendors that supply authoritative downstream implementation and signoff tools.

Across these scenarios, two boundaries matter: what broader platform vendors can absorb economically and what design organizations can justify building internally.

Standalone markets can emerge between them, but public evidence doesn't yet show who captures the value.

5.3 When Scarcity Moves

When technology makes one activity abundant, scarcity can move. As capabilities improve, costs fall, constraints shift, and organizational boundaries change, economic value can move to other parts of the agent–evaluator engineering loop. Proposal and exploration get cheaper, control splits across the loop, and technical capital accumulates in state and engineering memory.

Scarcity Migrates

Lower-cost generation is one way scarcity moves. A machine actor can propose more alternatives, revise them more often, and keep searching long after a human team would stop for lack of time or attention. As the candidate stream grows, the quality and cost of evaluation matter more because more evidence is needed to determine which candidates deserve further work.

The strategic value of high-authority evaluators can rise even if their costs don't fall. Lower-cost screening can reject obvious failures, allowing these evaluators to be used selectively for uncertain, promising, or consequential cases. Value comes both from the evaluator itself and from knowing when stronger evidence is worth the additional cost. Multi-fidelity optimization shows the routing pattern: inexpensive approximations broaden the search, and higher-cost checks are applied selectively when stronger evidence is needed. [19, 185, 186, 187]

If the cost of evaluation falls far enough, scarcity can shift again. A system may generate so many alternatives that, even after testing them, more candidates pass evaluation than an organization can responsibly compare and choose among. Automation shifts difficult residual work toward supervision, exception handling, and consequential decisions. [68, 84] Current engineering-agent accounts place people at requirements, checkpoints, validation, and evidence review. [72, 73] The scarce resource then becomes commitment: deciding when the evidence is sufficient, which tradeoff to accept, when exploration should stop, and who owns the consequences of that decision.

As broadly available AI models become more capable and less expensive, accumulated context becomes relatively more valuable. The more actions AI systems take without human intervention, the more important provenance and authorization become. As experimentation becomes easier and more frequent, failure history and routing policy become more valuable because they prevent the system from repeatedly paying to obtain evidence it already has.

AI can shift where engineering scarcity appears, but evidence, qualification, trust, and responsibility still shape what can be deployed, at what cost, and by whom.

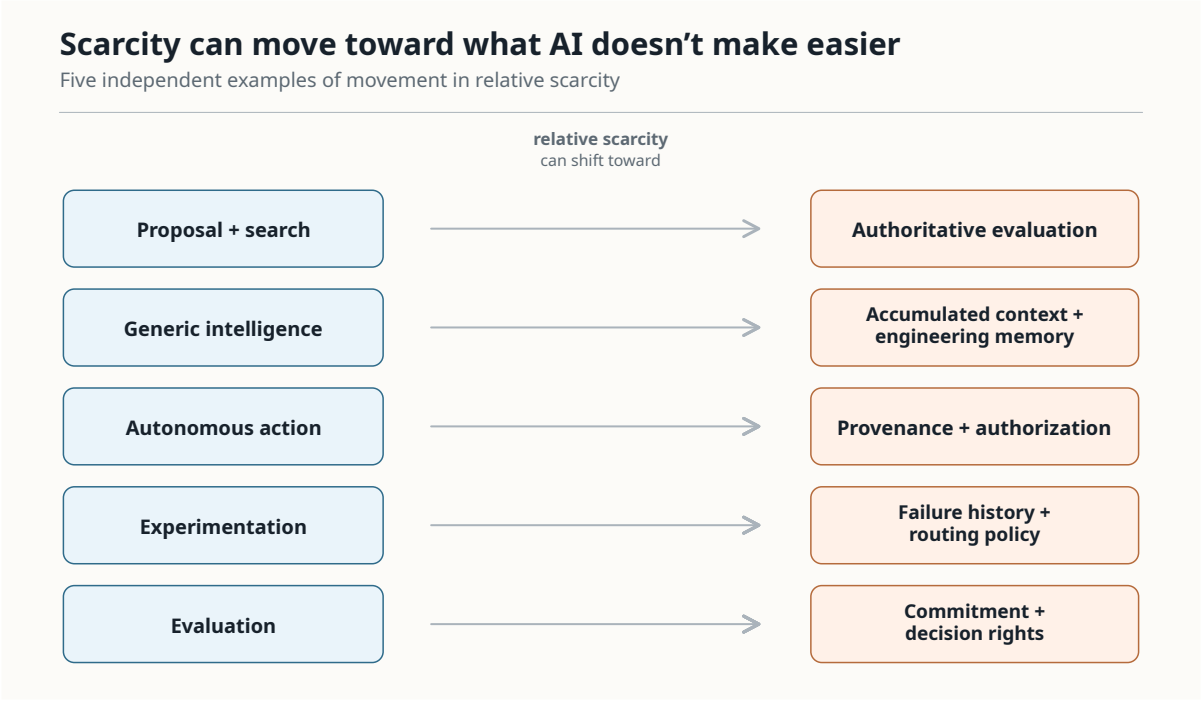


Figure 14. Reducing the cost of one engineering activity shifts relative scarcity to other parts of the engineering process.

Control, Dependence, and Economic Value Can Move Separately

Dimension	What it asks
Control	Who decides what happens next: who sets intent, controls state and routing, allocates budgets, orchestrates the workflow, and has authority to commit.
Dependence	Which capabilities the system can't readily do without. A design organization may control its workflow and still depend on external simulation, formal verification, implementation, signoff, circuit analysis, or other authoritative evaluators.
Economic value	Where economic value accumulates and who captures it. Control of the workflow doesn't determine who captures the value, and supplying an indispensable evaluator doesn't determine who controls the surrounding workflow.

A design organization can bring orchestration in-house while continuing to rely on external evaluation. Public examples already show organizations running proprietary design environments that combine commercial and internal tools, while hybrid arrangements keep organization-specific context separate from provider-owned evaluators and execution. [4, 5, 18, 183] Control over the workflow can move to the design organization without eliminating dependence on authoritative external

evaluators, and ownership of the surrounding technical capital doesn't have to follow either control or dependence. [65, 66, 77]

Strategic discussions often conflate control, dependence, and economic value in a single question: Who "owns the platform"? In AI-driven EDA, one actor may supply an indispensable capability, another may control the surrounding workflow, and economic value may accrue elsewhere.

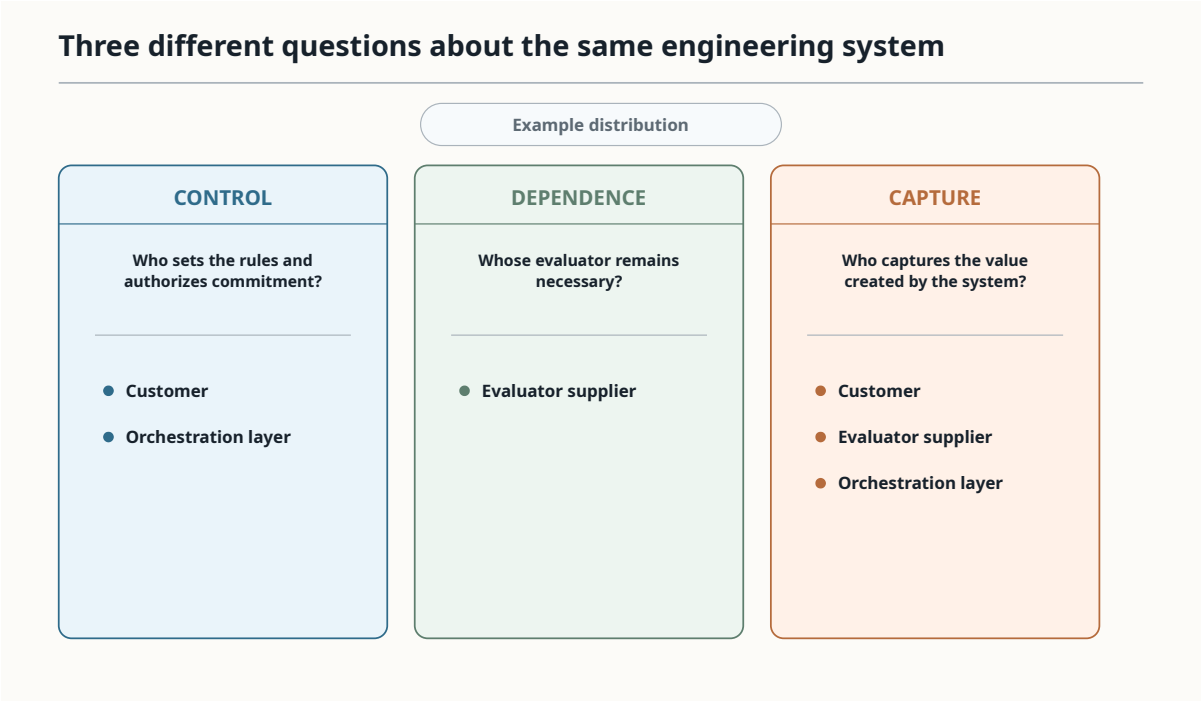


Figure 15. Control, dependence, and economic value can be distributed differently across the same system. A design organization can direct the workflow while relying on an external evaluator, with economic value distributed across several actors.

From Product Portfolios to Evaluator Portfolios

Portfolio	What it asks
Product portfolio	How capabilities are grouped into products; category growth, market size, margin, retention, competitive position, and sales of additional products to existing customers.

Portfolio	What it asks
Evaluator portfolio	What engineering question each capability can settle; how authoritative its answer is; what each evaluation costs in time, compute, license capacity, and organizational attention; which lower-cost checks can screen candidates before escalation; which evaluators can substitute for one another and which answer different questions; and when evidence must pass a qualification boundary before a decision can be made.

Research on algorithm selection and verification tools treats tool choice as a match among the task, required evidence, and resource constraints. [54, 55] An evaluator portfolio turns those choices into an engineering evidence architecture. Commercial packaging concerns how tools are sold and bundled.

A broad collection of engineering tools can become an evidence system rather than just a catalog when candidates can move from lower-cost screening to stronger evidence without losing state, provenance, and context along the way.

Portfolio breadth alone creates little advantage. Value comes from making authoritative evaluators, qualified integrations, persistent state, and routing policy work together as an evidence system. Technical capital accumulates in the qualified connections to those evaluators, the state preserved across handoffs, and the learned routing policy for choosing a useful path through the portfolio.

Routing Capital: What Evidence to Buy Next

An engineering system doesn't finish thinking when an evaluator returns a result. It needs to decide whether the result is enough, whether a different evaluator can settle the remaining question, whether stronger evidence is worth its cost, or whether the search should stop. Repeated evidence-routing decisions can become an organizational asset. Routing capital is accumulated, provenance-bound organizational capability for deciding what engineering evidence to buy next: which evaluator to invoke, at what fidelity, in what sequence, when to escalate, and when enough evidence has accumulated to stop. What counts as enough depends on the engineering question, the current design state, and the consequence of error. [51, 52, 53] Pyke uses the phrase "routing capital" slightly differently, to describe a firm's ability to decide where intelligence should happen. [59] We use it more narrowly for the organizational capability to decide what engineering evidence to buy next.

Computer science and engineering already route work among computational experts, models, algorithms, and levels of evaluation. [54, 55, 56, 57, 58]

Established tradition	Routing question
Mixture-of-experts systems	Which computational expert should receive an input?
Model routing	Which model offers the best expected quality for the cost?
Algorithm selection and verification portfolios	Which tool best fits the problem features and predicted performance?
Multi-fidelity optimization	When is a more expensive evaluation worth invoking?

Engineering evidence routing adds four dimensions that a simple destination choice can hide:

Engineering dimension	Routing consequence
Question fit	An evaluator can be decisive for one question and irrelevant to another.
Required design state	An evaluator can run only when the necessary state and assumptions exist.
Qualification boundary	Evidence carries authority only within the context in which it was qualified.
Decision authority	High fidelity alone doesn't give a result authority to settle the decision.

Engineering evidence routing is sequential and stateful and includes the choice to stop.

Routing capital forms when an organization learns from its evidence-routing choices. The router is infrastructure; the capital lies in validated organizational capability expressed across records, policies, integrations, and practices. A useful learning record preserves seven kinds of information:

Record element	What it preserves
Evaluator	Which evaluator was run
State and assumptions	What the evaluator saw
Cost and finding	What the evaluation cost and what it found
Decision effect	Whether the result changed the decision
Human intervention	When engineers overrode the route and what forced escalation
Stopping evidence	When additional evidence proved unnecessary
Validity context	Tool and PDK versions, design population, qualification history, failure record, and decision consequence

The learning record makes observations usable by tying each result to the state, assumptions, cost, evaluator, and decision effect that produced it. A routing policy is capitalized experience about what evidence is worth buying next. Routing capital improves productivity through five routing moves. [60, 62, 63]

How routing capital can improve productivity
Reject obvious failures with cheap evaluations.
Send uncertain or consequential cases to stronger evaluators sooner.
Order evaluations so early evidence changes later choices.
Avoid expensive runs unlikely to change the decision.
Stop when further evidence has little expected value.

Routing capital can compound. More evaluated decisions can improve the policy; better routing can make later evaluation more efficient; better-selected evaluations can produce cleaner learning signals; and those signals can improve the next route. Compounding depends on four validity conditions:

Validity condition	What must hold
Observed outcomes	The organization observes whether the evidence changed the decision, rather than merely recording that an evaluation ran.
Legible provenance	Each result remains tied to the state, assumptions, and evaluator that produced it.
Decision-linked success criteria	Success corresponds to the engineering decision the route was meant to support.
Change detection	The organization notices when tools, PDKs, design populations, evaluators, or methods change.

A silent evaluator defect, a biased design population, or a missing record of human overrides can turn experience into accumulated error. Activity alone doesn't create cumulative advantage. Routing capital compounds through the governed ability to distinguish valid experience from stale or misleading experience. [60, 61]

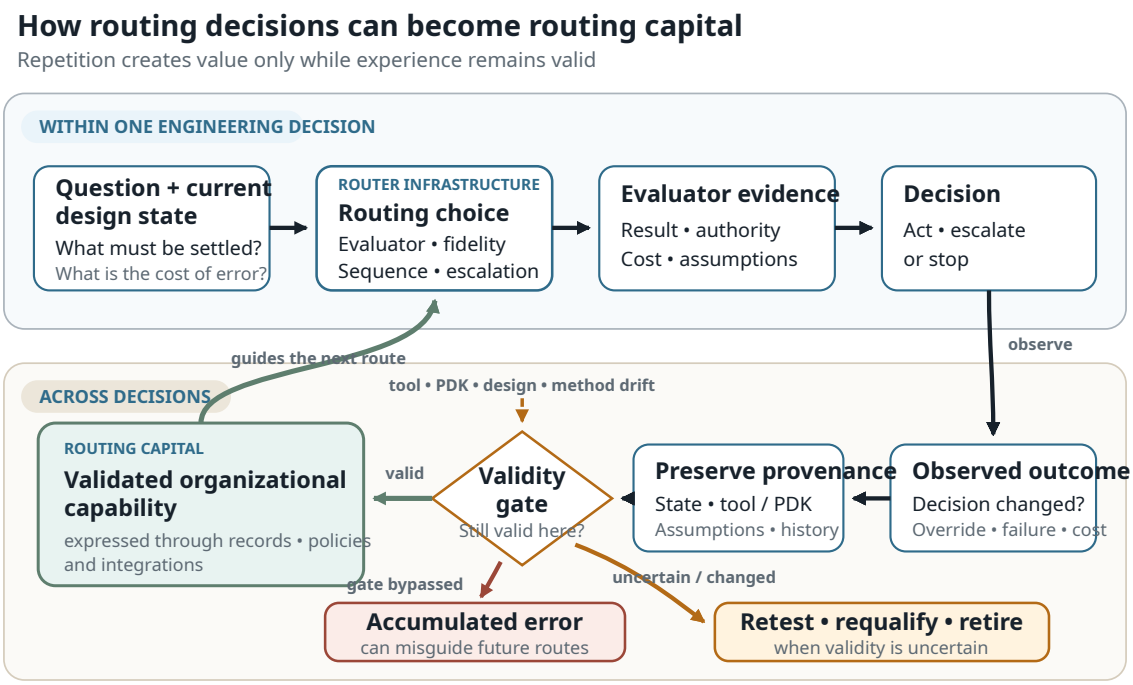


Figure 16. Routing capital connects two timescales. Within a decision, an engineering system routes the next evaluation and decides whether to act, escalate, or stop. Across decisions, observed outcomes become routing capital when provenance and validity checks preserve what remains reliable. Drift triggers retesting, requalification, or retirement. Experience that bypasses the validity gate accumulates error and can misguide later routes.

Some parts of routing capital are easier to transfer than others. Policy logic may move, while the evidence and operating context that made the policy reliable stay behind. A decision tree can be easy to export even when the design state, interfaces, cost structure, methodology, risk tolerance, evaluator behavior, and qualification history that gave it meaning remain local. Replacing an evaluator can require state translation, policy retraining, renewed evidence about failure modes, and requalification before engineers trust the new path's stopping and escalation rules. Five mechanisms distinguish what can move from what must be paid or relearned:

Mechanism	Question it answers
Transferability	What can move?
Context dependence	What loses value when separated from the organization, workflow, tools, data, and methods that made it useful?
Switching cost	What must be paid to change?
Requalification	What evidence must be regenerated before reliance is restored?
Path dependence	How do earlier choices constrain later ones?

EDA switching costs can arise when policy logic moves without its supporting context, when a new route must be requalified, or when earlier choices constrain later ones. [58, 60, 61, 64]

In EDA, routing capital is assembled from assets controlled by different parties:

Position	What it controls	What remains elsewhere
Design organization	Workflow direction	An authoritative signoff engine may remain with a vendor.
Platform provider	Route execution environment	The customer may retain the learning record and evaluation criteria.
Evaluator provider	A difficult-to-substitute evaluator	Other actors may retain the workflow, learning record, and design state.

Split control separates technical custody, legal ownership, workflow control, dependence, bargaining power, and value capture. Complementary assets, substitutability, qualification, and governance distribute control and value capture across the ecosystem. A hard-to-replace segment creates leverage; the surrounding system determines how much of that leverage becomes captured value. [65, 66]

Routing capital has a half-life, and its depreciation rate depends on the operating context. Managing depreciation requires an explicit response to each source of change:

Validity challenge	Response
Tool, PDK, library, or evaluator change	Retest affected routes and requalify evidence before relying on them.
New design population or failure mode	Re-establish where the policy predicts useful evidence.
Revised methodology or evidence requirement	Update stopping and escalation rules.
Rule that no longer predicts useful evidence	Retire it.

Stable decision logic can survive changes in tools, PDKs, libraries, evaluators, design populations, and methods; context-bound lessons expire when the conditions that produced them change. [60, 61]

Questions for assessing routing capital
What routing knowledge are we accumulating?
What proves that it remains valid?
Can we inspect and export the evidence history and routing record?
Who can use and update it?
What can move, what must be relearned, and what becomes stranded?
Who controls the capability, who depends on it, and who captures the return?

Answers to these routing-capital diligence questions reveal who governs the evidence, which capabilities travel, where dependence accumulates, and who captures the return.

Trust Across Ownership Boundaries

A design organization can retain proprietary design state, methodology, failure history, and engineering memory, and an external provider can supply an authoritative evaluator. Another layer may coordinate the interaction. Separately owned assets can still form one engineering system when the boundaries between them preserve what makes each asset valuable.

Across ownership boundaries, trust has a concrete architectural meaning. The system needs to establish:

Trust dimension	What must be established
Information flow	What information may cross the boundary
Authoritative state	Which state governs the interaction
Provenance	What record of origin and execution accompanies a result
Evaluator authority	Who may invoke a costly or sensitive evaluator
Commitment authority	Who may approve or commit a change
Reproducibility	How the result can be reproduced
Knowledge retention	What private knowledge the external party may not retain

Security and privacy are part of the boundary; authorization, evidence sufficiency, auditability, and ownership matter too. An autonomous system needs to support not only finding a promising answer but also establishing why the organization has authority to act on it.

An external provider can become more useful as the design organization's technical capital compounds, even though the provider doesn't own that capital. Richer private history can improve the questions sent to an evaluator; better methodology can improve routing; more complete state can make external evidence more relevant. The design organization can retain its most valuable private assets while getting more value from the external capability.

If the boundary lacks sufficient controls, the design organization may withhold context the system needs. If a provider can capture economic value only by owning the engineering memory, that ownership model may itself limit adoption. FedEDA is one example of a federated response to EDA privacy constraints. [11]

The mechanisms that let separately owned technical capital interact safely can become a strategic asset: qualifications, policies, interfaces, and accumulated confidence.

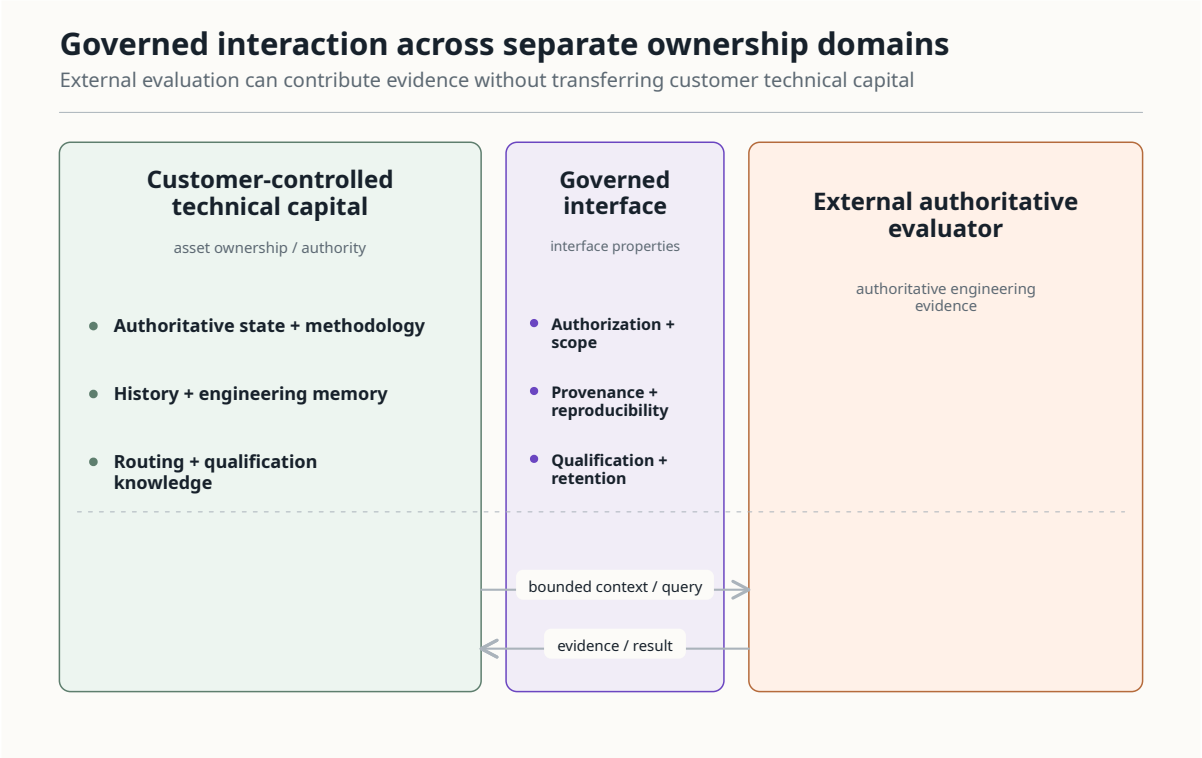


Figure 17. Separately owned technical capital can still operate as one engineering system through a governed interface. Bounded context can pass to an external authoritative evaluator and evidence can return without transferring ownership of the underlying capital. In the figure, 'customer-controlled' refers to ownership and authority over the technical capital, not to where it's deployed.

From Technical Capital to Economic Value

Over time, the important questions are what accumulates as a capability is used, who can retain that asset under current constraints, whether it transfers without losing value or leaking IP, how often it must be requalified, and whether it can support external pricing power. The same questions apply to products, internal engineering systems, acquisitions, partnerships, open infrastructure, and national capability.

Technical assets and arrangements create value through different mechanisms:

Asset or arrangement	Economic implication
Technically impressive capability	May not support durable standalone economics when its most valuable learning depends on proprietary context that can't be transferred or governed externally.
Ordinary integration	Can be strategically important when it preserves authoritative state across several difficult evaluators.
Acquisition	Can add functionality without adding a durable control point, while a qualified integration can strengthen several scarce assets at once.

Asset or arrangement	Economic implication
Reusable skill	May create large internal productivity gains without creating switching cost.
Trusted external evaluator	Can retain commercial power even when the design organization owns orchestration and engineering memory.

Technical importance, technical-capital formation, and commercial asset formation are three different things.

Capital-allocation questions
What knowledge, infrastructure, qualifications, routing policies, trust boundaries, or decision rights accumulate around that scarcity?
Which ownership structure do current conditions favor?
How transferable is it?
Who can turn it into durable economic value?

Following where scarcity moves can reveal the next layer of value even while ownership remains unsettled.

6. Diligence

6.1 From Demo to Decision-Grade Evidence

Engineering and capital decisions in AI-driven EDA depend on whether reported gains hold up in real evaluation and deployment. Impressive demonstrations can be especially easy to overinterpret.

Observed signal	What it doesn’t establish
A generated block compiles.	That the block implements the intended behavior correctly.
A PPA result is reported.	That the result beats a strong baseline under fair, comparable budgets.
A productivity claim is reported.	That the gain holds up through verification, cleanup, and the rest of the real workflow.
A customer logo is present.	Durable production deployment, repeated use, renewal, or switching cost.
A benchmark result is strong.	That the result holds up beyond the benchmark or that systematic failure modes are absent.

Observed signal	What it doesn't establish
A production tool has real customers.	That its strongest technical claim holds beyond the customers, designs, or conditions actually demonstrated.

Technical proof and commercial proof require separate evidence. Technology-readiness frameworks assess technical maturity. [85] Investment-readiness and customer-development frameworks assess adoption and business-model viability. [86]

Recent benchmarks make the early technical rungs more precise:

Benchmark	Capability tested
Phoenix-bench	Repository-scale hardware repair with executable feedback
ChipBench	Generation, debugging, and reference-model construction
PDAgent-Bench	Tool-centric physical-design execution
EDATracer	Evidence-grounded analysis across large artifact collections

Each benchmark defines a different capability, budget, and evaluator, so their scores shouldn't be collapsed into one autonomy measure. [104, 105, 106, 107]

Claim-specific verification sharpens this distinction: a successful endpoint score can conceal failed components or a bypassed verifier. [123] A recent practitioner account likewise distinguishes reaching GDS from repeatable convergence, signoff trust, and measured learning. [136] The account supplies contemporaneous framing rather than independent benchmark evidence.

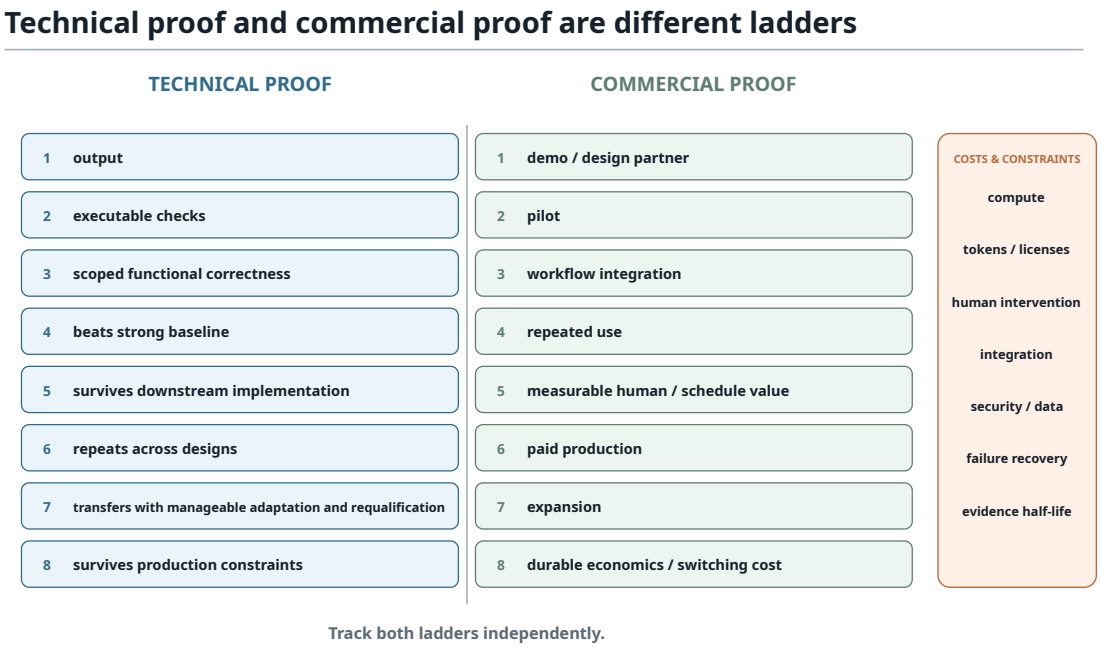


Figure 18. Technical and commercial success follow different evidence ladders. Commercial adoption can precede evidence that technical results generalize broadly.

Rung	Technical evidence	Commercial evidence
1	Generates output.	Demo or design partner.
2	Passes executable checks.	Pilot.
3	Is functionally correct within the tested scope.	Integrated into a real workflow.
4	Beats a strong baseline under fair, comparable budgets.	Repeated use.
5	Holds up through downstream implementation.	Measurable savings in engineering time or schedule.
6	Works across multiple designs.	Paid production.
7	Works in new settings without extensive retuning or requalification.	Expansion across teams or projects.
8	Holds up under production constraints.	Durable economics or switching cost.

Neither ladder substitutes for the other.

Example	What the evidence establishes	Next uncertainty
Rule2DRC	Executable evaluator with an explicit benchmark. [181]	Whether organizations adopt it for repeated paid production use.

Engineering memory	Trace2Skill establishes verifier-guided later-task improvement within its task regime; Dr. RTL establishes fold-isolated cross-design reuse; ChipMEM establishes bounded frozen-library transfer; DSO.ai adds provider-reported related-node reuse. [10, 20, 91, 98]	Durability, negative transfer, maintenance burden, cross-tool and cross-node portability, and rights to reuse across organizations.
DSO.ai + Cerebrus	Strong production/tapeout evidence. DSO.ai also has a company-reported related-node retargeting example in which reuse reduced the Fusion Compiler workload sixfold versus cold start. [1, 2, 91] Vendor-reported performance still requires scope-aware interpretation.	Whether the reported gains hold beyond the demonstrated scope and translate into durable economic value.
ChipStack	Named enterprise deployment and expansion evidence. [3]	Whether customers renew and whether meaningful switching costs emerge; that information remains private.
ChipAgents	Named customer workflows, bounded quantitative results, reported expansion beyond evaluation, more than 200 reported deployments, and customer-controlled deployment and data boundaries. [6, 145, 146, 147, 148, 149, 150, 151, 152, 153]	Independent verification, renewal and expansion, service effort per deployment, telemetry and update arrangements, ownership and reuse of learned artifacts, requalification burden, and switching behavior.

As a capability climbs the ladders, the decisive uncertainty shifts: first whether it can work, then whether it holds up in real workflows, then whether customers use it repeatedly and expand its use, and finally who captures durable economic value. The ladders become decision-useful only when the costs and constraints are tracked alongside the evidence: compute, licenses and tokens, human intervention, integration burden, security and data dependence, failure recovery, and evidence half-life.

As the decision-critical uncertainty shifts, so does the evidence worth buying next. The most informative next evidence is the experiment that resolves the most decision-critical uncertainty per dollar.

Claim under diligence	**Most informative next evidence**
Young verification agent	Production regression study with hidden bugs.
Higher-level representation	Downstream QoR on a real block.
Engineering-memory claim	A preregistered durability and negative-transfer study across a new tool, process node, or organization, using frozen memory and explicit requalification rules.

Claim under diligence	Most informative next evidence
Control-plane company	Evidence that customers expand usage after initial integration.
Physical surrogate	Correlation with signoff results on out-of-distribution designs.

The test is whether the organization made better engineering decisions faster, with acceptable risk.

AI systems can now climb the early evidence rungs rapidly, which increases the importance of the evaluator and evidence system. Durable value comes from repeatedly converting experiments into trusted decisions. Information about retention, churn, deployment burden, and procurement remains largely private.

6.2 What Evidence Would Change the Map?

The economic importance of evaluator allocation depends on search costing much less than authoritative evaluation. Whether that condition holds depends on specific technical and commercial factors.

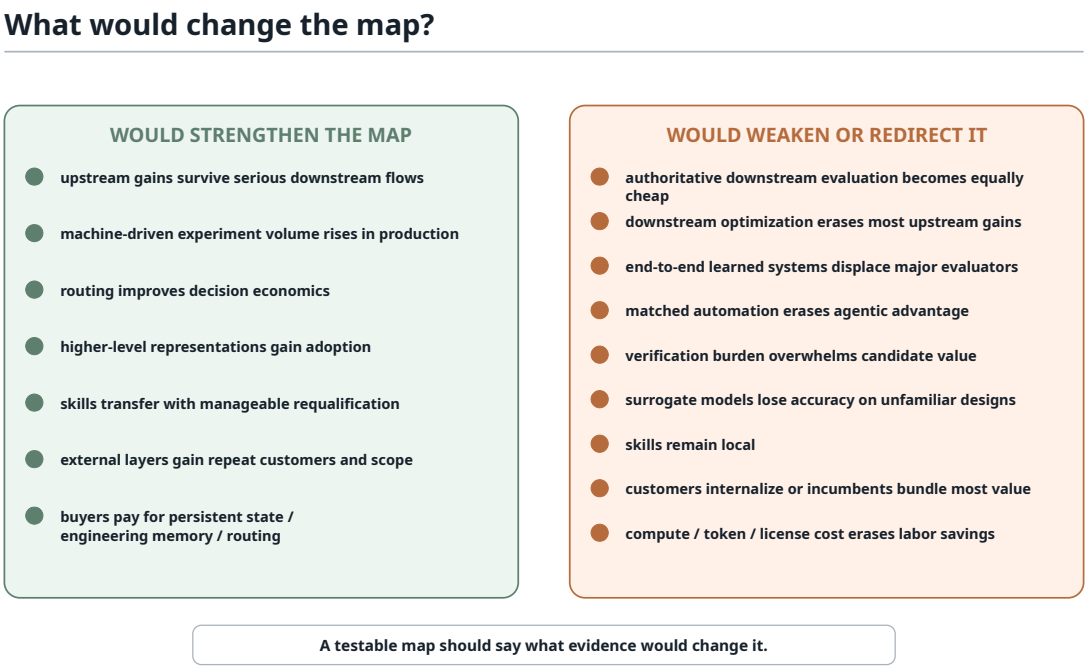


Figure 19. Evidence can strengthen, challenge, or redirect the proposed architecture and market interpretation.

Evidence that strengthens the map
Upstream gains repeatedly hold up through strong downstream commercial flows.
Agent-driven experiment volume increases substantially in production.
Higher-level representations are adopted more widely because AI lowers the cost of using them.
Engineering skills transfer across tools, process nodes, or organizations with bounded negative transfer and manageable requalification.
Neutral orchestration vendors gain repeat customers and expand their role within customer workflows.
Multi-fidelity evaluators maintain strong correlation with signoff results as design distributions change.
Customers pay for persistent workflow integration and engineering memory, not just access to the model.
Machine-driven tool use produces measurable economic value after accounting for token, compute, and license costs.

Evidence that challenges or redirects the map
The cost of authoritative downstream evaluation falls enough that the upstream/downstream cost asymmetry largely collapses.
Upstream alternatives that look better early no longer show meaningful advantages after strong downstream optimization.
The cost of verifying additional candidates consistently exceeds the value gained from generating them.
End-to-end learned systems repeatedly displace major downstream evaluator functions instead of only orchestrating them.
AI doesn’t substantially reduce the verification and maintenance work that’s limited higher-level representations.
Learned engineering skills don’t transfer well enough, or require too much revalidation, to be reusable across projects.
The organizational cost of choosing among additional acceptable alternatives consistently outweighs the value gained from broader search.
Token, compute, and license costs equal or exceed the value of the engineering effort saved.
Valuable control-plane capability is mostly internalized by design organizations or incorporated into incumbent platforms, leaving little durable market for independent providers.

Evidence needed to assess durable value capture
Customer renewal and churn.
Customer procurement decisions and rationale.
Ownership of persistent engineering memory across organizational boundaries.

Evidence needed to assess durable value capture
Implementation and ongoing services burden.
How industry-wide spending responds as costs fall.
Customer willingness to pay for cross-vendor neutrality.
Detailed dollars-per-truth curves across major commercial evaluator classes.

Public evidence can tell us a great deal about what's becoming possible, how new architectures are taking shape, and how technical control is being distributed. Evidence on renewal, procurement, switching costs, deployment burden, and willingness to pay addresses a different question: who actually captures durable value. Answering that question requires evidence closer to actual commercial behavior: management interviews, customer references, procurement data, and direct operating metrics.

7. Conclusion — Where Scarcity and Value Move

Chip design was already highly automated before the current AI wave. What's changed is the cost of generating, revising, and exploring alternatives. A persistent computational actor can propose alternatives, invoke tools, interpret evidence, remember outcomes, revisit failures, and keep searching at a volume a human team couldn't practically sustain.

Lower proposal and exploration costs expand what's practical when human effort in authoring, coordination, interpretation, or repetitive search is the limiting factor. Their effect is smaller where QoR requirements, approximation limits, solver complexity, process dependence, or implementation constraints dominate.

The cost of proposing and exploring alternatives can therefore fall faster than the cost of obtaining authoritative engineering evidence. That asymmetry puts more weight on evaluation decisions: what evidence to buy next, when stronger evidence is worth its cost, when the evidence is sufficient, and when to stop or escalate. More intelligence can move above and between mature engines without making those engines dispensable.

Control of the resulting agent–evaluator engineering system can be distributed among multiple actors. Different actors can set goals, maintain authoritative state and engineering memory, control evaluator access, choose routing policy, allocate budgets, and decide what to commit to. Technical capital can accumulate across design organizations, incumbent platforms, independent providers, open infrastructure, and sovereign capability. Technical importance and standalone marketability are different properties. Independent markets are most durable when a capability is valuable enough to buy, difficult to internalize, and difficult for an incumbent to incorporate economically into a broader platform.

As search and evaluation throughput rises, scarcity can move again toward commitment. Someone still needs to decide when the evidence is sufficient, which tradeoff to accept, when exploration should stop, and who bears the consequences if the choice is wrong.

Human judgment doesn't disappear; it moves toward decisions that require someone to take responsibility.

Public evidence already supports several parts of this shift: agentic systems can operate above trusted EDA engines, control can be distributed across the loop, and state, engineering memory, routing knowledge, and qualified integration can become technical capital. That evidence doesn't yet settle how ownership will be structured over the long run. Incumbent-provider, design-organization, independent-provider, hybrid, and open-infrastructure arrangements can coexist.

Cheap search does not eliminate scarcity. As search gets cheaper, different constraints can limit the system, different assets can accumulate around those constraints, and technical and economic value can concentrate in different places. For anyone trying to understand where the EDA ecosystem is headed, the task is to track those shifts: what's getting cheaper, what's becoming relatively scarcer, which capabilities and assets accumulate around the new limiting constraint, and what could cause value to move somewhere else.

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Appendix A — Company and Capability Map

This appendix is a reference map, not a fixed taxonomy. “Ownership tendency” records the structure most consistent with the current public evidence and current conditions; both the evidence and the conditions can change.

INCUMBENT AGENT / OPTIMIZATION PLATFORMS

Entity	Category / mechanism / evaluator	Ownership tendency / maturity / evidence	Durable assets / structural significance
Synopsys	Category: incumbent optimization, agent and orchestration platform, and hybrid specialized-model layer. Mechanism: DSO.ai uses reinforcement learning for design-space optimization [87, 88]; Copilot and AgentEngineer support assistant and agentic workflows; Autopilot orchestrates long-horizon agents; GPT-Synopsys is designed to operate Synopsys tools and iterate toward verified outcomes. [69, 72, 92, 94] Evaluator: Synopsys implementation/signoff engines. [89, 90]	Ownership tendency: Synopsys controls its tools and evaluators. GPT-Synopsys adds joint model development and OpenAI hosting, while customers may connect their own agent harnesses and engineers review outcomes. [92, 93] Deployment maturity: DSO.ai is at production scale. GPT-Synopsys has been announced, with early technology engagements underway. [1, 92, 179] Evidence: Synopsys reported the first 100 commercial DSO.ai tape-outs by February 2023, production use in about 20% of its top 100 Place & Route customers as of the end of 2023, named customer engineering gains, and about 20% uplift versus baseline Fusion contract value. [1, 179]	Competitive / structural considerations: DSO.ai benefits from close integration with Synopsys implementation/signoff infrastructure. Synopsys also reports bounded learned-state reuse: a related-node retargeting example reduced Fusion Compiler workload sixfold versus cold start. [91] GPT-Synopsys extends tool integration into a specialized model while permitting customer-harness interoperability. [92, 93] Orchestration owned by the design organization can still shift control toward the surrounding workflow. Commercial characterization: company-shaped.

Entity	Category / mechanism / evaluator	Ownership tendency / maturity / evidence	Durable assets / structural significance
Cadence	Category: incumbent optimization and agent platform. Mechanism: Cerebrus, ChipStack, Verisium/JedAI, native simulation/formal/P&R execution. Evaluator: Cadence engines.	Ownership tendency: incumbent or hybrid. Deployment maturity: Cerebrus production scale; ChipStack early enterprise expansion. Evidence: more than 1,000 Cerebrus tapeouts reported; named production customers; more than ten early ChipStack customers reported in 2026. [2, 3]	Competitive / structural considerations: control plane owned by the design organization; mixed-vendor flows. Commercial characterization: company-shaped.
Siemens EDA	Category: incumbent agent/data/orchestration platform. Mechanism: EDA-specific data layer, specialized parsers, RAG, tool automation, multi-model operation, self-verifying agents. Evaluator: Siemens deterministic/physics engines and connected tools.	Ownership tendency: incumbent or hybrid. Deployment maturity: productized architecture; publicly available deployment detail is more limited. Evidence: strong primary architecture statements. [15]	Competitive / structural considerations: mixed-vendor environments; preference within design organizations to retain control of institutional memory. Commercial characterization: company-shaped.

INDEPENDENT AGENT / WORKFLOW VENDORS

Entity	Category / mechanism / evaluator	Ownership tendency / maturity / evidence	Durable assets / structural significance
ChipAgents	Category: independent front-end design, verification, debug, and workflow agents. Mechanism: specification, RTL, UVM, stimulus, simulation-orchestration, debug, coverage, and root-cause agents over customer EDA environments. Evaluators: simulation, regression, coverage, formal, timing, design-rule, foundry-rule, and golden-simulation checks. [145, 146, 147, 148, 151]	Independent provider, commonly deployed in organization-controlled environments. Named Andes, eMemory, Whalechip, and Ambiq workflows report bounded results; Ambiq expanded beyond evaluation; ChipAgents reports more than 200 semiconductor-enterprise deployments as of October 2026. Renoir supports on-premises and air-gapped deployment, with customer control of design data and security policy. [145, 146, 147, 148, 149, 150, 153]	The provider can accumulate integrations, orchestration patterns, deployment methods, and support experience; customers can retain workflows, acceptance criteria, audit trails, and evidence about trust. Renewal, service burden, telemetry, derived-artifact ownership, requalification, cross-customer reuse, and switching behavior remain diligence questions. Commercial characterization: company-shaped; long-run durability remains open.
Bronco AI	Category: verification/debug/regression automation. [26] Mechanism: agent playbooks over regression analysis, waveforms, tickets, and debug workflows. Evaluator: simulator/regression environment.	Ownership tendency: independent or hybrid. Deployment maturity: production use is publicly described; detailed named-customer evidence isn’t yet widely public.	Competitive / structural considerations: native verification suites may increasingly incorporate comparable workflow automation. Commercial characterization: plausibly company-shaped.
Rise Design Automation	Category: higher-level representation and AI-operated design. [27] Mechanism: structured design representation plus deterministic generation/evaluation. Evaluator: synthesis and external EDA, including higher-fidelity physical feedback.	Ownership tendency: independent provider or design organization. Deployment maturity: real-project and production-design use is publicly described; detailed customer evidence isn’t yet widely public.	Competitive / structural considerations: design organizations may build related generator/HLS capabilities internally; incumbent front ends and open compiler infrastructure provide alternative paths. Commercial characterization: plausibly company-shaped; the public record is still early for judging long-run commercial structure.

Entity	Category / mechanism / evaluator	Ownership tendency / maturity / evidence	Durable assets / structural significance
Agentrys	Category: agent-workforce infrastructure for chip design. [28] Mechanism: engineering agents owned and improved internally by the design organization.	Ownership tendency: independent enabler of a design-organization control plane. Public deployment evidence: current cited sources provide limited detail.	Competitive / structural considerations: long-run differentiation may depend on semiconductor-domain integration beyond general platform/API capabilities and on preferences within design organizations for internal ownership. Commercial characterization: public evidence is still early for judging long-run commercial structure.
Chipmind	Category: neutral orchestration / RTL-agent environment. [29] Mechanism: tool-agnostic execution around existing flows.	Ownership tendency: independent. Public deployment evidence: current cited sources provide limited detail.	Competitive / structural considerations: long-run differentiation may depend on accumulating durable workflow state, integration depth, domain knowledge, or other assets that are difficult to reproduce. Commercial characterization: public evidence is still early for judging long-run commercial structure.

WORKFLOW / SYSTEM-OF-RECORD INFRASTRUCTURE

Entity	Category / mechanism / evaluator	Ownership tendency / maturity / evidence	Durable assets / structural significance
Defacto Technologies (Siemens acquisition announced)	Category: upstream SoC integration, persistent design database, design automation. [13, 30] Mechanism: design-state database plus SoC construction/integration and implementation-aware analysis.	Ownership tendency: independent provider; Siemens announced an agreement to acquire Defacto Technologies in July 2026. [30]	Strategic significance: concrete example of upstream workflow/data infrastructure being integrated into a major EDA portfolio. Commercial characterization: company-shaped; acquired.

Entity	Category / mechanism / evaluator	Ownership tendency / maturity / evidence	Durable assets / structural significance
DSM Pro	Category: programmatic design/project state. [31] Mechanism: metadata for build flow, tools, licenses, compute, storage, scripts, metrics, and reports.	Ownership tendency: independent provider or design organization.	Competitive / structural considerations: existing internal scripting and workflow infrastructure; design-organization preference to own state. Commercial characterization: plausibly company-shaped; detailed public commercial evidence remains limited.
Innova Advanced Technologies	Category: workflow/project orchestration. [32] Mechanism: machine-addressable project/flow management.	Ownership tendency: independent/hybrid.	Strategic significance: supports control-plane thesis. Commercial characterization: capability- or company-shaped, depending on integration depth.

PHYSICAL / ANALOG / SURROGATE

Entity	Category / mechanism / evaluator	Ownership tendency / maturity / evidence	Durable assets / structural significance
Astrus	Category: analog physical-design automation. [33] Mechanism: RL/search grounded in circuit/physics simulation. Evaluator: high-fidelity circuit simulation and eventual silicon.	Ownership tendency: independent.	Competitive / structural considerations: process/node generalization; incumbent analog tools; organization-specific data. Commercial characterization: plausible vertical company.
Alsemy	Category: semiconductor modeling / surrogate AI. [34, 191] Mechanism: learned compact/device models and simulation acceleration. Evaluator: calibrated physical/simulation data.	Ownership tendency: independent provider or foundry/design-organization hybrid.	Commercial characterization: plausible vertical company.
Good Intelligence	Category: combinatorial optimization for semiconductor physical design. [35, 192] Mechanism: optimization of macro placement and related chip-design choices.	Ownership tendency: independent provider.	Public evidence establishes an active product direction; deployment scale and comparative performance remain unverified.

HIGHER-LEVEL / STRUCTURED DESIGN INFRASTRUCTURE

Entity	Category / mechanism / evaluator	Ownership tendency / maturity / evidence	Durable assets / structural significance
XLS / DSLX	Category: open HLS/compiler infrastructure. [36]	Ownership tendency: shared/open or internal platform.	Strategic significance: deterministic machine-actionable foundation for higher-level design. Commercial characterization: infrastructure/capability-shaped, not standalone.
Chisel / SpinalHDL / Amaranth	Category: hardware construction/generator languages. [193, 194, 195]	Ownership tendency: ecosystem or design organization.	Strategic significance: parameterized design spaces can become attractive machine action spaces. Commercial characterization: infrastructure/capability-shaped.
ITDA Semiconductor	Category: structured SoC generation. [37] Mechanism: high-level/visual intent to RTL, UPF, SDC, and collateral.	Ownership tendency: independent.	Potential durable asset: encoded domain knowledge and generator methodology. Commercial characterization: plausible vertical company.
SoC Planner	Category: architecture exploration to synthesizable RTL. [182] Mechanism: A-DECA/VPSim + SoC construction + workflow/project layer.	Ownership tendency: public/industry hybrid.	Strategic significance: strong example of “search upstream, export into trusted conventional flow.” Commercial characterization: strategic capability; ownership fragmented.

Entity	Category / mechanism / evaluator	Ownership tendency / maturity / evidence	Durable assets / structural significance
Dr. RTL	Category: agentic RTL optimization / reusable skills. Mechanism: tool-grounded closed-loop optimization plus learned pattern–strategy library.	Ownership tendency: research, design-organization, or provider possibilities remain open. Evidence: a 47-entry pattern–strategy library evaluated on 20 real-world RTL designs. In each of four folds, skills were extracted from 15 designs and applied to five unseen designs. The full system reports average improvements of 21% in WNS, 17% in TNS, and 6% in area relative to an industry-leading commercial synthesis tool. [20]	The test establishes bounded cross-design reuse; cross-tool, cross-node, cross-organization, and long-horizon validity remain open. Commercial characterization: technical capability demonstrated; a commercial model isn't yet established by the cited public evidence.

ENGINEERING MEMORY / REUSABLE-SKILL SYSTEMS

Entity	Category / mechanism / evaluator	Ownership tendency / maturity / evidence	Durable assets / structural significance
Trace2Skill	Category: engineering-memory / reusable-skill research system. Mechanism: repeated rollouts on difficult Verilog tasks are distilled into a task-specific natural-language skill using sanitized verifier feedback. Evaluator: task verifier. [10]	Ownership tendency: research or design-organization capability. Evidence: verifier-guided later-task improvement within its task regime; frozen-library, cross-design, cross-tool, and cross-organization transfer weren't tested.	Strategic significance: shows that evaluated traces can become an explicit reusable skill. Commercial characterization: research capability; ownership and business model remain unresolved.

Entity	Category / mechanism / evaluator	Ownership tendency / maturity / evidence	Durable assets / structural significance
ChipMEM	Category: verification-grounded engineering-memory research system. Mechanism: procedural memories admitted after synthesis, simulation, or formal checks, with evolving and frozen libraries tested on tasks that weren't used for initial library development. [98]	Ownership tendency: research or design-organization capability. Evidence: With a frozen procedural memory library, ChipMEM produced accepted testbenches for all 20 CVDP tasks that weren't used to build the library, versus 18 of 20 without a procedural memory library; it also reports 39 of 54 versus 35 of 54 equivalence-passing outputs overall. Each configuration was evaluated once.	Strategic significance: establishes bounded frozen-library transfer. Durability, negative transfer, maintenance, cross-tool portability, and cross-organization reuse remain open. Commercial characterization: research capability; ownership and business model remain unresolved.

OPEN / SHARED INFRASTRUCTURE

Entity	Category / mechanism / evaluator	Ownership tendency / maturity / evidence	Durable assets / structural significance
OpenROAD	Category: open RTL-to-GDS implementation and evaluator infrastructure. [38]	Ownership tendency: ecosystem.	Strategic significance: lowers the cost of experimentation, benchmarking, and agent integration. Commercial characterization: infrastructure-shaped; value may accrue above or around it.
iEDA / AiEDA	Category: open EDA and AI-oriented infrastructure. [16, 42, 156, 201, 202]	Ownership tendency: shared ecosystem.	Strategic significance: open foundation for AI-driven EDA experimentation and integration of tools and agents in China. Commercial characterization: infrastructure-shaped.
CIRCT	Category: hardware compiler IR infrastructure. [39]	Ownership tendency: ecosystem.	Strategic significance: structured intermediate representation and compiler transformations. Commercial characterization: infrastructure-shaped.

Entity	Category / mechanism / evaluator	Ownership tendency / maturity / evidence	Durable assets / structural significance
OpenFASoC	Category: generator ecosystem. [40]	Ownership tendency: ecosystem.	Strategic significance: reusable generators as technical capital. Commercial characterization: infrastructure-shaped.

DESIGN-ORGANIZATION / HYBRID EXAMPLES

Entity	Category / mechanism / evaluator	Ownership tendency / maturity / evidence	Durable assets / structural significance
Google internal Silicon Design Environment	Category: internal design-organization control plane. Mechanism: a coherent environment of first- and third-party tools, Google-specific and standard interfaces, scalable flows, shared methodology, and procedural and AI-first automation. Evaluators: commercial and internal tools provide question-specific evidence. [5]	Ownership: design organization. Current role materials establish development and deployment responsibility across the environment; they don't report realized productivity or ROI.	Direct evidence that a design organization can control workflow, state, interfaces, and methodology while remaining dependent on external evaluators. Reuse can compound internal value; tool and node changes create requalification burden.
Apple internal CAD / Signoff AI / RTL construction	Category: internal design-organization control plane. Mechanism: signoff dashboards and metrics, a Signoff AI assistant, custom implementation and signoff tools, ML/GenAI methods, RTL-construction automation, documentation, training, and direct work with EDA vendors. Qualified commercial and internal checks supply the decision evidence. [18, 183, 184]	Ownership: design organization. Current role materials establish broad internal capability building; they don't report realized productivity, cost, or portfolio-wide ROI.	Internal methodology, metrics, interfaces, and operating knowledge can compound through reuse while external engines and vendor expertise remain essential. Changes in tools, nodes, and methodology create requalification burden.
NVIDIA + Cadence	Category: hybrid control plane. Mechanism: organization-specific engineering context and AI infrastructure combined with incumbent execution/evaluation.	Ownership tendency: shared/hybrid.	Strategic significance: shows ownership can split cleanly across the loop. [4]

NATIONAL / PUBLIC INFRASTRUCTURE

Entity	Category / mechanism / evaluator	Ownership tendency / maturity / evidence	Durable assets / structural significance
QiMeng / Chinese Academy of Sciences	Category: national-lab automated hardware/software design. [41, 196, 197]	Ownership tendency: public research/national capability.	Strategic significance: strategically valuable capability without a conventional software-market requirement.
CEA A-DECA / VPSim / SoC Planner	Category: French public/industry architecture-exploration systems and programs. [13, 182]	Ownership tendency: public/industry hybrid.	Strategic significance: Established architecture-exploration infrastructure can become more useful as AI reduces the expert effort needed to operate it, extend it, and explore alternatives with it.
Japanese national GenAI+HLS/local-model programs	Category: national capability programs. [14]	Ownership tendency: mixed public/private.	Strategic significance: sovereignty, workforce leverage, and design capability are explicit economic objectives beyond software revenue.

Company-shaped versus capability-shaped is an analytical distinction, not a score or a permanent label. The characterization reflects current evidence about purchase behavior, integration depth, and where compounding assets can be retained; it can change as those conditions change. A capability can create very high enterprise or national value without supporting a durable standalone vendor.

Appendix B — Evidence Summary for Major Claims

The tables below connect each major claim to its public evidence and to the boundary on the conclusion. The References section provides the complete cited sources.

Claim	Evidence state	Public evidence and boundary
Claim 1 — Agent-over-trusted-engine architecture	SUPPORTED by public evidence.	Cadence ChipStack, Siemens self-verifying agents, Synopsys's public progression from DSO.ai optimization through Copilot and AgentEngineer to Autopilot and the announced GPT-Synopsys model/tool architecture, Rule2DRC, and Spec2RTL-Agent. [1, 3, 15, 69, 70, 72, 92, 94, 181] Agents can propose or orchestrate while established tools execute and validate. The evidence supports plural ownership models.
Claim 2 — AI-driven EDA has entered real enterprise deployment	SUPPORTED by public evidence.	DSO.ai commercial tapeouts; Cerebrus production/tapeout evidence; ChipStack named enterprise deployment; and ChipAgents named quantitative workflows, reported expansion beyond evaluation, and more than 200 reported deployments. [1, 2, 3, 6, 145, 146, 147, 148, 149] Public evidence is much stronger on deployment and bounded workflow outcomes than on retention, churn, service burden, or long-run unit economics.
Claim 3 — The stack is economically asymmetric	SUPPORTED as an analytical model.	Foundry/signoff dependence, early physical-prediction work, architecture-exploration systems, and incumbent agent architectures support the model. Optionality generally declines and authority/cost of truth generally rise toward physical/manufacturing closure. Analog and cross-stage physical-aware design prevent a simple front-end/back-end wall.
Claim 4 — The new control plane supports plural ownership structures	SUPPORTED by public evidence.	Incumbent platforms; Apple and Google internal CAD; independent agent vendors operating over customer evaluators and organization-controlled deployment; the NVIDIA and Cadence hybrid; and the distributed model, tool, customer-harness, and engineer-review structure announced for GPT-Synopsys. [3, 5, 6, 15, 18, 92, 93, 150, 151, 153, 180] Public evidence establishes plural control structures; it doesn't establish which one captures the largest long-run value pool.

Claim	Evidence state	Public evidence and boundary
Claim 5 — Strategically valuable but non-vendible technical capital is real	SUPPORTED as a category.	Apple and Google internal methodology/platform investment; privacy/local-model constraints; proprietary design-state logic. [5, 11, 18, 183, 184] Internal value isn't directly measurable from public evidence.
Claim 6 — AI can make some previously constrained automation approaches more practical	SUPPORTED in narrowed form.	HLS studies document productivity gains alongside persistent QoR tradeoffs; SoC Planner, Japanese GenAI+HLS programs, and generator/DSE research show current efforts to automate exploration and higher-level design. [9, 13, 14] AI can reduce constraints arising from human authoring, search, and interpretation. AI doesn't erase QoR, fidelity, solver, or physics constraints.
Claim 7 — Evaluator economics is a practical engineering framework and shapes autonomy	SUPPORTED by public evidence.	Allocation lineage includes engineering information economics, multi-fidelity optimization, value of computation, and risk-informed verification and validation. [22, 23, 24, 25, 51, 67, 188, 189] EDA cases include Rule2DRC, formal/SMT, HLS, VPSim, and analog multi-fidelity optimization. [8, 9, 19, 181, 185, 186, 187] Evaluator economics treats evaluator choice, evidence sufficiency, stopping, and escalation as a practical engineering allocation problem. In EDA, cheap machine-readable judges can support larger search loops while stronger or more authoritative evidence is selectively escalated. The synthesis treats evaluator choice, evidence sufficiency, stopping, escalation, and authority across a set of evaluators as one practical engineering allocation problem. EDA evaluators don't form one universal scalar fidelity/authority ladder, and no universal commercial dollar-per-evaluation curve exists.
Claim 8 — Engineering memory is a real technical asset	SUPPORTED by public evidence.	Trace2Skill demonstrates verifier-guided later-task improvement within its task regime; Dr. RTL establishes fold-isolated cross-design reuse; ChipMEM establishes bounded transfer from a frozen library to tasks not used to build it; and a Synopsys-reported DSO.ai example demonstrates related-node warm-start reuse. [10, 20, 91, 98] The evidence establishes several bounded forms of reuse. Cross-tool, broader cross-node, cross-organization, and long-horizon validity remain open.

Claim	Evidence state	Public evidence and boundary
Claim 9 — Engineering memory creates a durable vendor advantage	CONDITIONAL / SCENARIO.	Skills must transfer, remain valid, and be legitimately retainable by the vendor across deployments. Case 5 shows that deployment and design data can remain customer-controlled while ownership of derived artifacts, telemetry, updates, and cross-customer reuse remains a contract and diligence question. [150, 153] Open questions include IP ownership, revalidation cost, negative transfer, cross-organization leakage, and memory controlled by the design organization.
Claim 10 — Neutral orchestration can become a major independent category	CONDITIONAL / SCENARIO.	Mixed-vendor flows and independent providers make the category plausible. Counterforces are incumbent platform integration and internalization by design organizations. Long-run pricing power and renewal remain open.
Claim 11 — Higher-level representation may become more valuable under AI	SUPPORTED as technical direction; CONDITIONAL as market outcome.	Agentic HLS, SoC Planner, Rise, and XLS/generator ecosystems. [9, 13, 14, 27, 50] Production adoption and external-vendor economics remain incomplete.
Claim 12 — More automation increases total EDA spending	CONDITIONAL / SCENARIO.	Agentic flows can raise evaluator and compute invocation volume. FlexEDA supports usage metering and budget controls, and industry participants report that token efficiency and budget predictability are becoming operational considerations. [4, 12, 21] Some workflows can also reduce expensive downstream runs. The net industry effect remains unsettled.
Claim 13 — Usage/compute pricing broadly replaces seat licensing	CONDITIONAL / SCENARIO.	By-the-minute EDA already exists. One metering model doesn't establish broad licensing transition.
Claim 14 — Multilingual evidence broadens the technical and ownership landscape	SUPPORTED by public evidence.	Looking beyond the sources most visible from Silicon Valley reveals technical capabilities and institutional structures across China, including both the People's Republic of China and Taiwan, France, South Korea, Japan, Germany, India, the United Kingdom, and Israel that are easy to miss from a U.S.-centered view. [16, 41, 42, 109, 110, 157, 159, 160, 161, 162, 167, 168, 181, 196, 197, 202, 204, 207, 208, 209, 210, 211] Coverage is broad, not exhaustive.

Claim	Evidence state	Public evidence and boundary
Claim 15 — Current production-oriented autonomy depends on governance	SUPPORTED by current public evidence.	Human-centered agentic verification, industry expert discussion, simulator truth checks, explicit guardrails, and the engineer-review and data-control structure announced for GPT-Synopsys. [92] Current evidence favors layered autonomy with human decision-making and strong evaluators. Future autonomy levels may change.

OPEN — PUBLIC EVIDENCE INSUFFICIENT

Seven decision-critical questions remain beyond current public evidence:

Decision-critical questions
Private procurement/build-vs-buy logic.
Renewal/churn/net retention for emerging vendors.
Customer-specific deployment services burden.
Cross-organization ownership of persistent engineering memory.
Industry-wide total-cost elasticity.
Detailed commercial cost/fidelity curves across evaluator classes.
Willingness to pay specifically for cross-vendor neutrality.

Evidence half-life

Half-life class	Evidence interpretation
FAST	Model benchmarks, agent feature scope, startup positioning, and model/token costs; treat conclusions as perishable snapshots.
MEDIUM	Customer deployments, licensing changes, workflow adoption, and company categories; useful for medium-lived conclusions about adoption and category shape.
SLOWER	Foundry qualification, the cost of authoritative physical evaluation, security and IP incentives, and established automation methods and infrastructure change relatively slowly, so evidence about them can support longer-lived conclusions.

Appendix C — Historical Cases: Which Constraints AI Relaxes

Architecture / case	Prior promise and limiting constraints	What AI relaxes	Limits AI won't remove / evidence-based assessment
High-level synthesis	Promise: Let designers describe algorithms at a higher level and have the compiler generate RTL. Blockers: Teams often have to restructure high-level code around the hardware the compiler can produce, learn how to steer the compiler with directives, debug results that can be hard to predict, and fit HLS into RTL-centered workflows. Even then, the resulting implementation can still trail expert RTL on QoR.	AI can reduce the expert effort required to restructure high-level code, explore directives, interpret reports, and keep the compilation loop running.	Limits AI won't remove: AI won't make every algorithm a good fit for HLS. Physical constraints still apply downstream, and the resulting implementation can still trail expert RTL on QoR. Evidence: Historical surveys report productivity gains alongside persistent QoR tradeoffs. Recent agentic-HLS research and Japanese programs apply AI to parts of the HLS workflow. [9, 14, 50] Assessment: AI can make HLS practical for more designs and teams, but it won't make every design a good fit for HLS or eliminate the QoR tradeoff with expert RTL.
Hardware construction languages and generators	Promise: Let designers capture reusable hardware structures in generators that can produce whole families of related circuits. Blockers: Teams have to learn a new language and ecosystem, debug generators whose mistakes can propagate widely, change methodology, and fit the approach into RTL-centered workflows.	AI can reduce the expert effort required to create and modify generators, explore parameter spaces, and translate design intent into reusable hardware structures.	Limits AI won't remove: Generator bugs can propagate widely, so generated designs still have to be verified and integrated into the downstream flow. Evidence: Chisel-based research uses generators for design-space exploration, XLS provides compiler infrastructure for higher-level hardware construction, and newer companies are pursuing structured hardware generation. Assessment: AI can reduce enough of the expert effort required to use hardware construction languages and generators to make them practical in a wider range of settings. Production-market evidence remains uneven.

Architecture / case	Prior promise and limiting constraints	What AI relaxes	Limits AI won't remove / evidence-based assessment
Virtual prototyping and HW/SW co-design	Promise: Use executable system models to test architectural choices and software behavior before committing to RTL. Blockers: Teams have to build and maintain the model, check it against later implementation results, often rely on specialists to operate it, and make sure its predictions are accurate enough to guide later decisions.	AI can reduce the expert effort required to maintain executable models, run workloads, interpret results, and keep experiments moving.	Limits AI won't remove: A fast model is only useful if it predicts the later implementation accurately enough to guide good decisions. Evidence: CEA's VPSim/A-DECA work and related software-defined-computing research use executable system models to explore architectural choices and software behavior before RTL. [182] Assessment: AI can reduce the effort of building and using executable models, but their predictions still have to be accurate enough to guide good decisions.
Analog expert systems and executable design knowledge	Promise: Capture expert analog design knowledge in executable rules and generators that can size circuits, choose among topologies, and produce design candidates. Blockers: Much of the design knowledge is tacit, good choices depend on the process and topology, high-fidelity simulation is expensive, and methods often don't transfer cleanly across topologies or process nodes.	AI can reduce the expert effort needed to encode design knowledge, generate candidates, search alternatives, and orchestrate simulator loops.	Limits AI won't remove: High-fidelity simulation remains expensive, process variation still matters, and methods may not generalize across topologies and nodes. Evidence: French CAIRO/OCEANE work encoded analog design knowledge into automated design systems; newer Japanese programs revisit executable analog knowledge; and reinforcement learning, Bayesian optimization, and surrogate models automate parts of circuit search. Assessment: AI can widen and accelerate analog search, but high-fidelity circuit evaluation, process variation, and generalization will still limit how far it can go.

Architecture / case	Prior promise and limiting constraints	What AI relaxes	Limits AI won't remove / evidence-based assessment
Design-space exploration	Promise: Explore many design alternatives systematically and compare them using explicit objectives and evaluator results. Blockers: The search system has to fit into the existing design flow, be usable by real engineering teams, pay for expensive solver and simulation runs, and produce enough value to justify adoption. A specialized provider also has to turn that capability into a viable business.	AI can reduce the expert effort required to orchestrate experiments, interpret results, choose the next action, and keep the search running.	Limits AI won't remove: Solver and simulation runs can remain expensive, and the system still has to fit into real engineering workflows. Evidence: Commercial systems including SoC Planner, DSO.ai, and Cerebrus, together with architecture-exploration research, use automated search and optimization to evaluate design alternatives. [2, 13, 87, 88] Assessment: AI can reduce the effort required to orchestrate experiments, interpret results, choose the next action, and keep search running.
Formal methods as a synthesis/repair foundation	Promise: Use formal specifications, proofs, and solvers to find design errors, verify properties, and sometimes generate or repair logic. Blockers: Teams have to write precise specifications, solvers can still struggle as problems scale, and counterexamples often require expert interpretation before they lead to a useful repair.	AI can reduce the expert effort required to write specifications, use formal tools, and interpret counterexamples.	Limits AI won't remove: Solver complexity remains, and proving that a design is wrong still doesn't tell you how to repair it. Evidence: Artificial Incorrectness found that formal checking can reliably identify incorrect generated designs, but giving counterexamples back to an LLM doesn't reliably produce a successful repair. [8] Assessment: AI can help with specification, tool use, and counterexample interpretation, but solver complexity and the difficulty of turning a counterexample into a good repair remain.

Appendix D — Evaluator Economics Table

Evaluator class	Relative cost/latency	Authority	Main use	Key limitation
Syntax / compile	Very low	Syntax/typing only	reject invalid artifacts	limited functional evidence
Simulation/tests	Low to medium	workload/test dependent	fast behavioral feedback	incomplete state-space coverage
Formal / SMT	Variable	high for specified properties	proof/counterexample	specification/scale/ repair gap
DRC	Low to medium	high for encoded rules	geometry/rule validation	rule scope only
HLS synthesis	Medium	strong for synthesizability, partial for final QoR	structured inner loop	QoR gap versus strong RTL
Learned surrogate	Very low after training	approximate	high-volume candidate ranking	drift/out-of-distribution risk
Virtual prototype	Low to medium	model-dependent	early HW/SW/architecture exploration	fidelity/calibration
Trial P&R / OpenROAD	Medium	stronger physical evidence	early physical filtering	less authoritative than production signoff
Full commercial P&R/signoff	High	high under qualified assumptions	production convergence	cost/latency/license burden
High-fidelity SPICE	High	high for circuit behavior/model	analog truth check	expensive sampling
Silicon	Extreme	empirical final behavior	ultimate validation	months and very high cost

Quantitative exemplars:

Example	Reported quantitative result
HLS	Historical surveys report roughly one-third development time and >4× productivity on average, with RTL QoR generally better. [9]
VPSim	Reported up to 600 MIPS for virtual-prototype exploration. [182]
Analog multi-fidelity	One 2026 study reports that its surrogate-guided mode replaces about 96.5% of SPICE calls and yields about 20× lower total simulation time while retaining selective SPICE checks. [19]

The hierarchy compares evaluator classes by relative cost, authority, use, and limitation.

Appendix E — Regional Evidence Coverage

The regional comparison tests whether the same technical architecture persists across different ownership, sovereignty, and commercialization environments. It compares architecture, ownership, and constraints across ecosystems.

Region / evidence base	Distinctive evidence or emphasis	What it makes visible
China — PRC	iEDA/AiEDA open EDA + AI infrastructure; QiMeng automated processor HW/SW; UniVista; Empyrean's separation of agentic design from authoritative simulation/signoff; multi-agent IC-design patents; Chinese-language SoC/PCB agent work.	Substantial domestic development spans open stacks, research systems, patents, and commercial platforms. [41, 42]
France	CEA A-DECA and VPSim; SoC Planner; Defacto; Innova; LIP6 CAIRO/OCEANE/Coriolis; substantial HLS, generator, and analog-automation research and tooling. AiEDA provides a current open-source data and interface layer, while Iskander, Louërat, and Kaiser provide older analog executable-knowledge lineage. [109, 114]	Some architectures now being developed with AI build on earlier work in executable design knowledge and design-space exploration. [182]
South Korea	Rule2DRC; KAIST AI-EDA Consortium; early physical prediction; POSTECH differentiable optimization/RL/analog/synthetic-data work; FedEDA; emerging startups.	The cost of truth and execution-grounded learning become especially visible. [11, 181]
Japan	Local-LLM-aided design; GenAI + HLS national programs; analog tacit-knowledge formalization; use-case-driven chip creation; JEDAT AI/analog direction; long HLS/generator history. The Okada program explicitly frames analog intent, constraints, and evaluation criteria as knowledge to be captured and inherited; it's a program objective rather than a completed outcome. [110]	Japan's current AI work builds on established HLS, generator, and design-automation infrastructure. Local deployment and security requirements also shape how these systems are built. [14]
China — Taiwan	NTU and MediaTek-linked research on AI-driven EDA; foundry-adjacent university/industry automation; startups; national semiconductor commercialization programs.	In Taiwan, AI-driven EDA is developing within a dense design/foundry ecosystem rather than only as standalone software. [43, 207]
Germany	Fraunhofer AI-assisted chip design; analog automation; verification/certification emphasis; industrial/public infrastructure.	Complementarity between AI and classical EDA is prominent; assurance and safety become capital considerations. [44, 208]
India	CaretEDA open-stack/agentic commercialization; large design-services/EDA engineering base; semiconductor-design policy and capability investment.	Open-stack commercialization and workforce/design-capability dimensions are especially visible. [45, 209]
United Kingdom	Proof-centric RTL-agent research; AI hardware/semiconductor capability programs; Arm-centered design ecosystem.	Proof/evidence-centric autonomy and sovereign design capability are prominent. [46, 210]

Region / evidence base	Distinctive evidence or emphasis	What it makes visible
Israel	Agentic-EDA research; vendor-neutral CAD/cloud service infrastructure; semiconductor startup/talent ecosystem; national AI/chip-design strategy.	Neutral integration can itself be a durable services/platform category even before agentic AI. [47, 211]
United States	Major EDA incumbents, NVIDIA, open-source infrastructure, AI-native startups, academic benchmarks, commercial reporting, and investor disclosures. Visible coverage emphasizes agentic platforms, verification startups, model benchmarks, OpenROAD/CIRCT/XLS, and incumbent product direction.	From a Silicon Valley vantage point, the sources easiest to encounter emphasize U.S. incumbents, startups, and research, while technical capabilities and institutional structures in European and Asian semiconductor ecosystems are easier to miss.

Cross-region synthesis

The regional evidence spans the United States; China, including both the People's Republic of China and Taiwan; France; South Korea; Japan; Germany; India; the United Kingdom; and Israel. Across these ecosystems, the evidence broadly supports a common architecture: proposal, search, and orchestration around trusted evaluators, with evidence and engineering memory guiding the next action. The important differences are ownership, openness, sovereignty, technical domain, evaluator type, and maturity.

Evidence reach and limits

Analytical dimension	Evidence reach / limit
Technical architecture	Across the regions examined, no separate dominant technical architecture emerged.
Ownership and capital capture	Coverage remains incomplete where the relevant evidence is private: public sources provide limited evidence on procurement, renewal, pricing, and internal platform economics.
Known limitation	Public-source coverage is broad, not exhaustive. Questions that depend on local practice, procurement, or nonpublic operating evidence require region-specific experts, customer references, management access, or other private evidence.

Appendix F — Method and Evidence Limits

Scope

The evidence base consists of public sources. The analysis focuses on decision-relevant uncertainties in technical mechanism, ownership, value capture, evaluator economics, deployment maturity, historical blockers, and falsifiers.

Evidence position and regional perspective

The author is embedded in the U.S. technology and EDA ecosystem and has greater direct familiarity with its institutions, companies, terminology, and English-language public record than with every other semiconductor ecosystem. Regional and multilingual research was used deliberately to reduce that asymmetry, not merely to broaden geographic coverage.

The source review reinforced the importance of looking beyond the sources most visible from Silicon Valley. Chinese sources revealed a dense domestic ecosystem of AI-driven EDA research, infrastructure, and commercial activity. French sources revealed substantial capabilities in architecture exploration, executable design knowledge, HLS, analog automation, and related methods whose current importance may exceed their commercial visibility. Other regions revealed different combinations of technical capability, institutional structure, sovereignty, public investment, and commercialization.

Recent surveys and technical literature

Literature area	Representative sources
Agentic EDA maps	[95, 96, 97]
Verification-grounded memory and tool-interactive execution	[98, 99]
Architecture and deployment evidence	[103, 108]
Hardware-agent benchmarks and artifact analysis	[104, 105, 106, 107]
Regional and historical EDA sources	[109, 110, 111, 112, 114]
Prediction and judgment economics	[102, 113]
Validation congestion and evaluator design	[116, 117, 118, 123, 133, 140, 141, 142]
Transferable skills, memory, and harnesses	[115, 119, 120, 121, 122, 124, 125, 126, 127, 128, 129, 130, 131, 132, 137, 138, 139]
Bounded physical-design agents	[143, 144]
Commercial and practitioner context	[134, 135, 136]

Source classes

Included source classes
Peer-reviewed papers
Preprints
Patents
Primary company documentation
Named customer case studies
Conference talks and technical webinars
Public investor disclosures
Government and national-program material
Repositories and benchmarks
Job postings
Independent industry reporting
Archived technical/project material
Multilingual regional sources
Recent surveys and review articles
Roadmaps, vision and decadal-agenda material
Workshop and community research reports
Standards outlooks and active standards projects
Major institutional research programs and program histories

Evidence states

Evidence state	Meaning
SUPPORTED	Public evidence is sufficient to support the claim within its stated scope.
CONDITIONAL / SCENARIO	The mechanism or direction is credible, but the evidence doesn't justify a categorical economic conclusion.
OPEN — PUBLIC EVIDENCE INSUFFICIENT	Relevant facts are private, undisclosed, or not observable well enough to conclude. An open question marks the point where the available evidence is insufficient and a different kind of evidence is needed.

Interpreting commercial evidence

Evidence type	What it establishes	What it doesn't establish
Vendor press release	What the vendor is claiming and where it's placing a product or capability.	Independent performance; that requires independent measurement or corroboration.
Named customer case	A relationship crossed a meaningful threshold.	Renewal, expansion, switching behavior, or durable economics.
Funding	Capital formation around a category.	Product-market fit without sustained use, expansion, renewal, or other durable customer behavior.
Customer logo	A relationship exists.	Production use or depth of deployment without more specific evidence.
Benchmark	Capability within the benchmark's design, baseline, and budget.	Production maturity under representative constraints.
Job posting	Where a company is investing in internal capability.	ROI or procurement logic, which remain on the private side of the evidence boundary.

Evidence half-life

Evidence half-life	Typical evidence and interpretation
FAST	Model benchmarks, frontier-model capability, startup feature scope, and token/model pricing. Treat conclusions built on these as perishable snapshots.
MEDIUM	Customer deployments, agent-product integration, licensing changes, startup categories, and build-vs-buy patterns. These support medium-lived conclusions about adoption and category shape.
SLOWER	Foundry qualification, the cost of authoritative physical evaluation, security and IP incentives, established automation methods and infrastructure, and process/foundry integration change relatively slowly. Evidence about them can support conclusions beyond individual model and product cycles.

Evidence selection and coverage

Evidence-selection principle	Purpose / criterion
Broad source coverage	Different parts of the AI-driven EDA transition become visible in different source classes.
Multilingual coverage	Helps uncover semiconductor capabilities and institutional structures that are less visible from a Silicon Valley vantage point.
Counterevidence and alternatives	Includes counterevidence, failure modes, and alternative architectures so vendor and research results are interpreted against competing explanations.
Inclusion threshold	A distinct architecture, ownership structure, mechanism, quantitative result, or failure mode, or material corroboration of an existing finding.

Public-evidence boundary

This report draws its evidence from the public record and uses no management interviews, customer-reference calls, expert-network engagements, proprietary procurement data, or confidential operating metrics.

The public-evidence boundary shows where public evidence stops and private economics begin. Procurement, renewal, pricing, switching cost, internal-platform economics, and other nonpublic facts can materially change a capital decision. In this analysis, the closer the question gets to durable economic advantage, the more of the decisive evidence moves behind the public boundary.

Private evidence required
How customers actually make procurement and build-vs-buy decisions.
Whether deployments renew, expand, or churn.
How much services burden deployment carries.
When design organizations will externalize persistent design memory.
Who owns learned skills and whether they transfer across organizations.
How industry-wide demand responds to total cost.
What commercial evaluator cost/fidelity curves look like across classes.
What internal platforms actually return on investment.

Appendix G — Survey Guide to AI and Automation in EDA

AI and automation in EDA span several technical lineages. The surveys and reviews below provide entry points into LLM and agentic systems, established automation methods, and analog design.

LLM and Agentic Approaches to EDA

Literature pathway	Survey or review	Language and publication status	Coverage and use
Broad LLM-for-EDA map	王静 et al.,《大语言模型在电子设计自动化中的应用综述》 [163]	Chinese; peer-reviewed survey	Maps RTL, HLS, backend, analog, and workflow automation; foregrounds tool-chain integration, correctness, security, data, IP, and privacy.
Agentic EDA system architecture	林薇 et al.,《智能体驱动的电子设计自动化技术与应用》 [164]	Chinese; national-platform preprint	Organizes agentic EDA around foundation models, knowledge augmentation, tool calling, multi-agent collaboration, and human control.
Agent handoffs	Liu et al., “Agentic Electronic Design Automation: A Handoff Perspective” [95]	English; arXiv preprint	Explains producer–consumer handoffs, hierarchical feedback, persistent context, and acceptance evidence.
Autonomous digital-chip design	Zang et al., “The Dawn of Agentic EDA” [96]	English; arXiv preprint	Surveys autonomous digital-chip design from front-end work through physical design and cross-stage feedback.
RTL generation and benchmarking	Ravindran et al., “Survey and Benchmarking of Large Language Models for RTL Code Generation” [17]	English; preprint	Entry point for RTL-generation methods, benchmark structure, and open evaluation problems.
Broad LLM research map	Pan et al., “A Survey of Research in Large Language Models for Electronic Design Automation” [172]	English; peer-reviewed survey	Compact map of model architectures, customization methods, design stages, integration challenges, and research opportunities.

Literature pathway	Survey or review	Language and publication status	Coverage and use
Critical LLM-for-EDA synthesis	He et al., “Large Language Models for EDA: Future or Mirage?” [171]	English; peer-reviewed survey	Longer benchmark-aware synthesis across generation, verification and debugging, knowledge retrieval, and optimization.
Early hardware-design overview	Xu et al., “LLM-Aided Efficient Hardware Design Automation” [176]	English; arXiv overview	Concise 2024 map of HDL generation, repair, verification, and physical implementation with two case studies.
Expanded cross-workflow overview	Xu et al., “Large Language Models (LLMs) for Electronic Design Automation (EDA)” [177]	English; conference-accepted paper available as arXiv preprint	Broadens the earlier overview through a larger author group and case studies in design, testing, and optimization.
Front-end roadmap	Xu, Li, and Schlichtmann, “LLM for EDA in Front-End Design” [178]	English; invited DAC paper available as arXiv preprint	Focused path from specification understanding through HDL and testbench generation, HLS, and agentic execution.
Japanese industry view	小田川 and 牧井, 「AI の適用による半導体設計の効率化・自動化の取り組みと現状」 [169]	Japanese; vendor-authored professional review	Provides a concise practitioner view of AI-assisted digital design and Cadence Cerebrus, offering industry context rather than independent performance evidence.
Independent EDA providers and startup landscape	Jesse Allen, “EDA Startups At DAC 2025.” [190]	English; trade ecosystem survey, 2025	Maps verification agents, project-state infrastructure, higher-level design, and SoC-generation startups; use as a discovery map, then follow the first-party company sources.

Established EDA automation lineages and multi-fidelity optimization

Literature pathway	Survey or review	Language and publication status	Coverage and use
HLS adoption and outcomes	Lahti et al., “Are We There Yet? A Study on the State of High-level Synthesis” [9]	English; peer-reviewed study	Synthesizes development-time, productivity, and QoR evidence across 46 papers and 118 applications.
HLS mechanisms and adoption history	若林, 「ソフトウェアプログラムからハードウェア記述を合成する高位合成技術」 [167]	Japanese; peer-reviewed review	Explains HLS mechanisms and the engineering work required to move from prototype systems to practical tools.
FPGA HLS languages and systems	三好, 「FPGA 向けの高位合成言語と処理系の研究動向」 [168]	Japanese; peer-reviewed survey	Maps languages and processing systems that raise FPGA design above RTL and reduce development and verification burden.
Machine learning across FPGA EDA	田 et al., 《基于机器学习的 FPGA 电子设计自动化技术研究综述》 [166]	Chinese; peer-reviewed survey	Covers machine learning in FPGA HLS, logic synthesis, placement, and routing as a pre-agentic automation lineage.
Intelligent global and detailed routing	周 and 卓, 《芯片智能布线方法研究综述》 [165]	Chinese; peer-reviewed survey	Maps planning, search, and learning methods, evaluation, datasets, tools, and open problems in chip routing.
Shift-left and cross-stage correlation	Wu et al., “Shift-Left Techniques in Electronic Design Automation” [173]	English; arXiv preprint	Connects earlier prediction and virtual design environments to downstream behavior; makes proxy fidelity and adoption timing explicit.
Multi-fidelity optimization	Li and Li, “Multi-Fidelity Methods for Optimization: A Survey” [23]	English; peer-reviewed survey	Provides the adjacent methods for allocating work across evaluators with different costs and fidelities.

Analog design and layout

Literature pathway	Survey or review	Language and publication status	Coverage and use
ML-assisted analog design	Takai, “Automated Design of Analog Integrated Circuits Using Machine Learning” [170]	English original; invited peer-reviewed paper	Focused technical entry point into machine-learning-assisted analog topology and parameter design.
Human–AI collaboration across analog design	Poddar et al., “Artificial Intelligence-Assisted Analog IC Design” [174]	English; peer-reviewed magazine synthesis	Maps tightly coupled decisions across the analog flow and the complementary roles of machine search and human design intent.
Automated analog layout and open infrastructure	Berhan et al., “Automated Analog Layout Synthesis” [175]	English; peer-reviewed open-access review	Explains ALIGN, designer constraints, automated layout generation, and layout-in-the-loop optimization.